



Ch #1

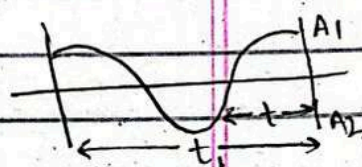
Signal:-

have a varying component
(as a function of time or space)
which conveys information

through variations e.g. electric
charge density. When an electric
charge is distributed
over a body, the signal is the charge
density which is a function of space rather than
time.

Size of a signal:-

must consider not only
the signal amplitude but also
its duration.



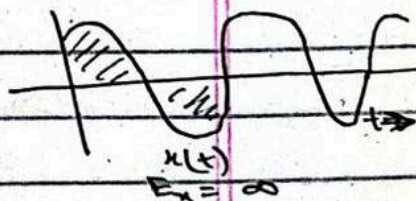
Signal Energy:-

Area under the signal $g(t)$ is
a possible measure of its size.

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt \rightarrow \text{for real signal}$$

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt \rightarrow \text{for complex signal}$$

If $g(t) \not\rightarrow 0$ as $t \rightarrow \infty$ signal
energy is infinite.



Signal Power:-

Defined for periodic signals

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt \rightarrow \text{for real}$$

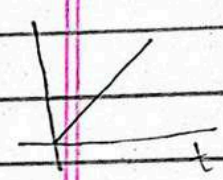
$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt \rightarrow \text{for complex}$$

$\Rightarrow$ Periodic signals $g(t) = g(t + T_0)$ for all T .

A signal which repeats itself after certain time is called periodic signal.

~~$x(t)$~~ $x(t) = t \rightarrow$ Ramp signal.

Here Signal energy $= \infty$
 Signal power = periodic signal
 $=$ not found



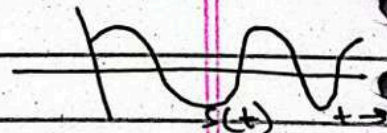
Classification of Signals:-

1) Analog Signal:-

An analog signal is any continuous time ~~and~~ continuous amplitude signal.

$$s(t) \rightarrow t \in \mathbb{R}$$

For mathematical treatment, $s \in \mathbb{C}$

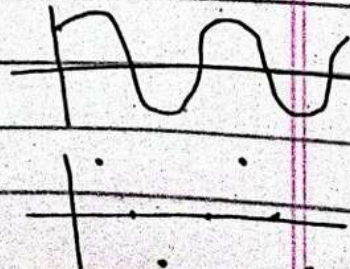


2) Discrete Time Signal:-

These are signals designed on a discrete time domain $\{nT_s | n \in \mathbb{Z}\}$ and assume continuous values.

$$s(nT_s) \in \mathbb{C}, n \in \mathbb{Z}$$

Discrete signals are derived by sampling at instants that are multiples of T_s .



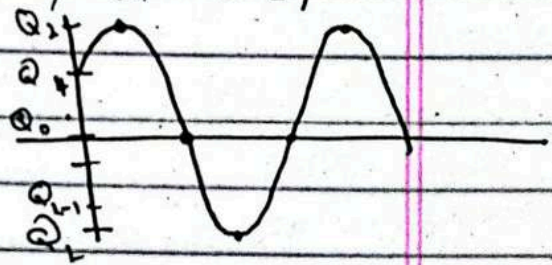
3) Digital Signal:-

A digital signal defined on a discrete time domain $\{nT_s | n \in \mathbb{Z}\}$ and is discrete in both time and amplitude. So, a digital signal satisfies

$$S(nT_s) \in A, n \in \mathbb{Z}$$

where $A = \{Q_0, Q_1, Q_2, \dots, Q_L\}$

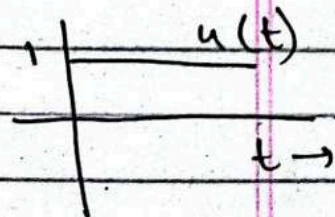
with A as set of L elements



Useful Signal Models:-

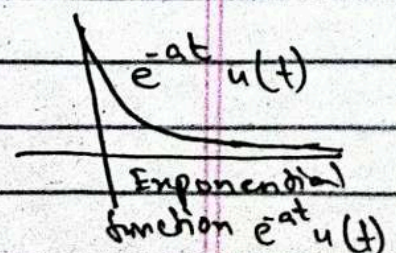
1) Unit Step function $x(t)$

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$



$\Rightarrow$ Physical example

The unit step function can be represented by closing of a switch at time $t=0$ if a battery of V_0 voltage is connected to the network at $t=0$.

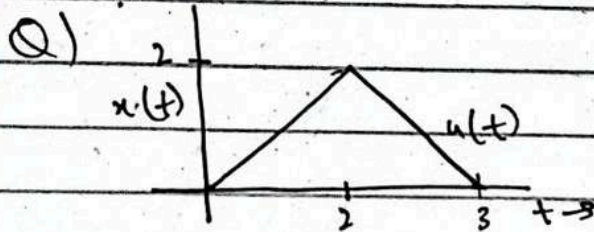
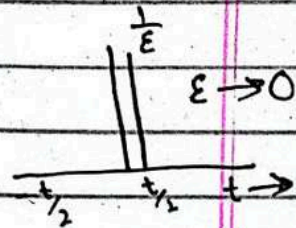
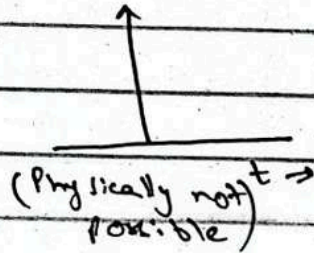


$$\int \phi(t) \delta(t-\tau) dt = \phi(\tau)$$

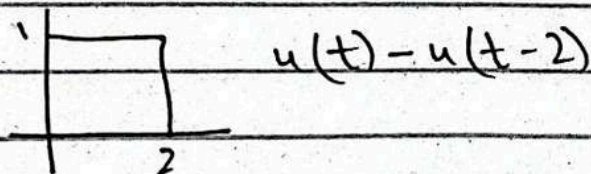
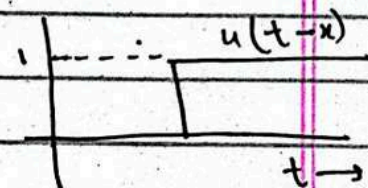
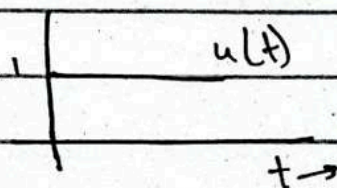
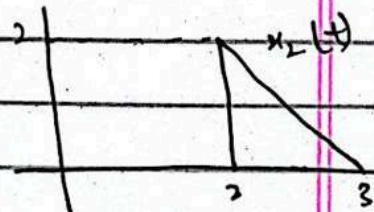
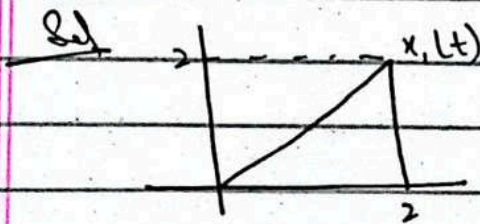
Unit Impulse function

$$\delta(t) = 0, t \neq 0$$

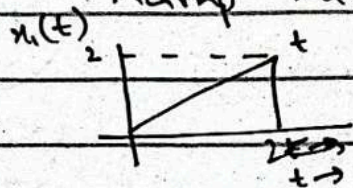
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$



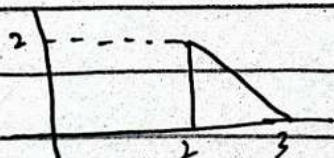
Write function of $u(t)$



Ramp Function:-



$$x_1(t) = t[u(t) - u(t-2)]$$



Triangular pulse $[u(t-2) - u(t-3)]$

The required ramp has slope of -2 and is thus $-2t + c$.

Ramp has 0 at $t=3$, so the constant $c=6$. Thus the ramp is $-2(t-3)$.

$$x_2(t) = -2(t-3)[u(t-2) - u(t-3)]$$

$$x(t) = x_1(t) + x_2(t)$$

$$= t[x(t) - u(t-2)] + [-2(t-3)[u(t-2) - u(t-3)]]$$

$$= t[x(t) - x(t-2)] - 2(t-3)[x(t-2) - x(t-3)]$$

$$= tu(t) - tu(t-2) - 2tu(t-2) + 2u(t-3) + 6u(t-3) - 6u(t-3)$$

$$= tu(t) - 3tu(t-2)$$

$$= t[u(t) - u(t-2)] - 2(t-3)[u(t-2) - u(t-3)]$$

$$= tu(t) - tu(t-2) - 2(t-3)u(t-2) + 2u(t-3)(t-3)$$

$$= tu(t) - 3tu(t-2) - 2tu(t-2) + 6u(t-2)$$

$$+ 2tu(t-3) - 6u(t-3)$$

$$= tu(t) - 3tu(t-2) + 6u(t-2) + 2tu(t-3)$$

$$- 6u(t-3)$$

A.

$$= tu(t) - 3u(t-2)[t-2] + 2u(t-3)[t-3]$$

1) Analog Signal

$$s(t), \quad t \in \mathbb{R}, \quad s \in \mathbb{C}$$

2) Discrete Signal

$$s(nT_s) \in \mathbb{C}, \quad n \in \mathbb{Z}$$

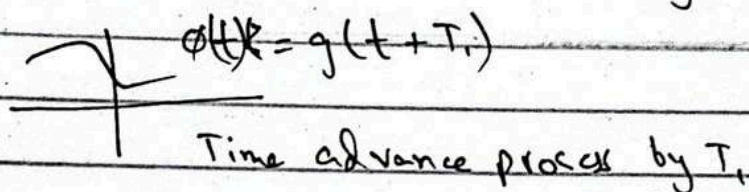
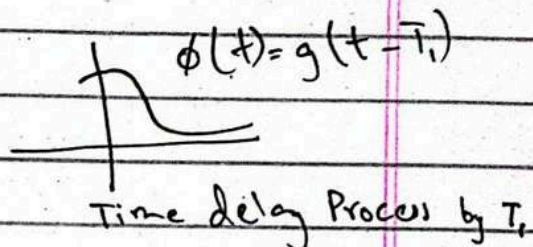
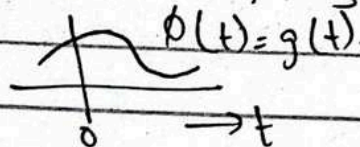
3) Digital Signal

$$s(nT_s) \in A, \quad n \in \mathbb{Z}$$

$$A = \{Q_0, Q_1, Q_2, \dots, Q_n\}$$

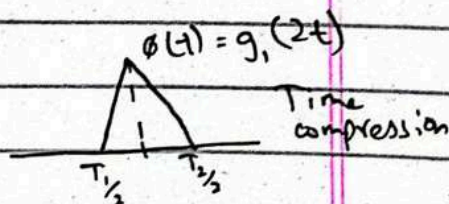
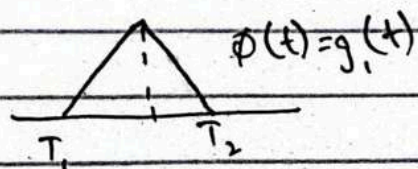
Signal Operations:-

1) Time shifting

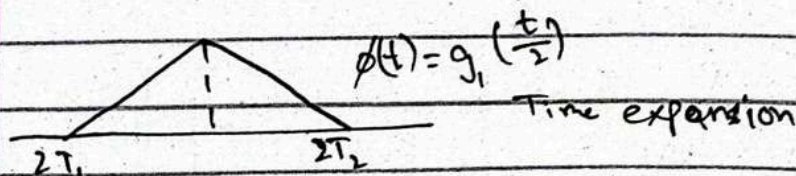


A signal $x(t)$ delay by T seconds.
It is expressed as $x(t - T)$, advanced
by T is $x(t + T)$

2) Time Scaling

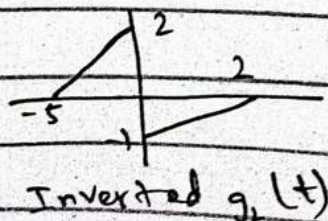
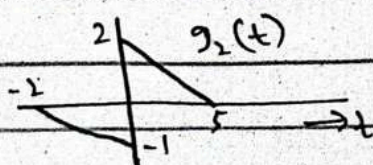


$g(t)$ time compressed by factor 'a' is $g(at)$



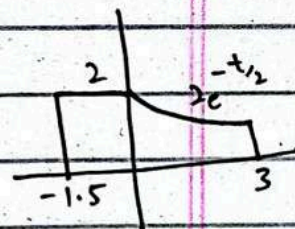
$g(t)$ time expanded by factor 'a' is $g(\frac{t}{a})$

3) Time Inversion



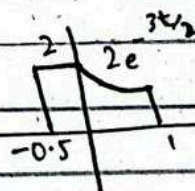
Q) Sketch & describe mathematically $x(3t)$

$$x(t) = \begin{cases} 2, & -1.5 \leq t \leq 0 \\ 2e^{-t/2}, & 0 \leq t \leq 3 \\ 0, & \text{otherwise} \end{cases}$$



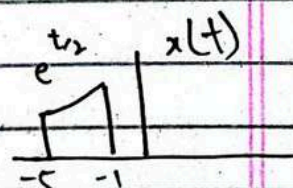
$x(t)$ needs to be compressed by a factor of '3'

$$x'(t) = x(3t) = \begin{cases} 2, & -0.5 \leq t \leq 0 \\ 2e^{-3t/2}, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

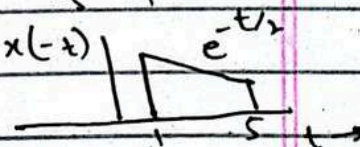


Q) Time reversal

$$x(t) = \begin{cases} e^{t/2}, & -1 \leq t \leq 5 \\ 0, & \text{otherwise} \end{cases}$$



$$x'(t) = \begin{cases} e^{-t/2}, & 1 \leq t \leq 5 \\ 0, & \text{otherwise} \end{cases}$$



Systems:-

A system processes input signal to modify them or extract additional information from them to produce output signal (response).

A system maybe made up of physical components, hardware realization or it maybe an algorithm that computes an output signal from an input signal.

Software Realization:-

The study of systems consists of three major areas: mathematical modeling, analysis and design.

Basic System Properties:-

1) System with and without memory:-

A system is said to be memoryless if its output for each value of the independent variable at a given time is dependent only on the input at that time.

$$\text{e.g. } y[n] = (2x[n] - x^2[n])^2$$

⇒ A resistor is a memoryless system, input current, output voltage, the input output relation is $y(t) = Rx(t)$

⇒ A discrete-time system with

memory is an accumulator or summer.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \rightarrow y[n] = \sum_{k=-\infty}^n x[k] \times x[n]$$

$$y[n] = \underbrace{y[n-1]}_{\text{memory}} \times x[n]$$

⇒ Capacitor is a continuous time system

with memory.

$$y(t) = \frac{1}{C} \int_{-\infty}^{\infty} x(\tau) d\tau$$

(where C is capacitor)

2) Invertibility and Inverse Systems:-

A system is said to be invertible if distinct inputs lead to a ~~the~~ distinct output.

e.g. $y(t) = 2x(t)$

for which the inverse system is

$$w(x) = \frac{1}{2} y(t)$$

3) Causality:-

A system is causal if the output at any time depends only on the values of the input at the present time and in the past.

A causal system is one for which the output at any instant t_0 depends only on the value of the input $x(t)$ for $t \leq t_0$.
 $y(t)$ at $t_0 \Rightarrow$ input $x(t)$ for $t \leq t_0$

So the output depends only on past and present values and not on future values. The causal system are also those

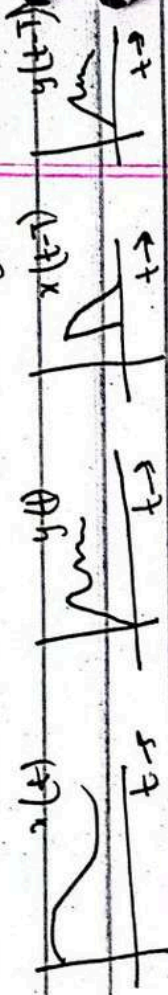
called physical or non-anticipative system.

In simple words, in the ~~causal~~ causal the output cannot start before input is applied.

4) Stability:-

Bounded input, bounded output
Systems are termed as stable systems.

5) Time Invariant and Time Varying Systems:-



$$x(t) \Rightarrow y(t)$$

$$x(t-t_0) \Rightarrow y(t-t_0)$$



The systems and the time delay
delay commute if the system is
time-invariant.

A system is time invariant
if a time shift in input signal
results in an identical time shift
in output signal.

Linear Systems:-

A linear system possess property of superposition. The system is linear if

1. The response of $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$ (additivity)
2. The response of $ax_1(t)$ is $ay_1(t)$ where 'a' is any complex constant (homogeneity/scaling)

Superposition = Additivity + homogeneity

Linearity implies homogeneity (scaling) and additivity

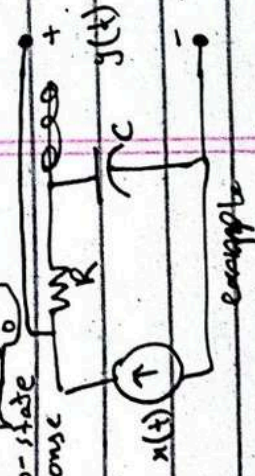
$$x_1 \rightarrow y_1 \quad ax_1 \rightarrow ay_1$$

$$x_2 \rightarrow y_2$$

$$K_1 x_1 + K_2 x_2 \rightarrow K_1 y_1 + K_2 y_2$$

The equation of this circuit

$$y(t) = \underbrace{y_c(0)}_{\text{zero input response}} + \underbrace{\frac{1}{C_0} \int x(\tau) d\tau}_{\text{zero state response}}$$



Linearity allows decomposition which permit separation of an output into components.

Linear & Non-linear Systems:-

Almost all systems observed in practice become non-linear when large enough signals are applied to them. However, it is possible to approximate most of the non-linear systems by linear systems for small signal analysis.
Going ahead

1) Superposition property greatly simplifies the analysis of linear systems.

→ Zero-input component is suppressed and zero-state component is computed by ~~also~~ assuming zero-initial condition.

$$x(t) = a_1 x_1(t) + a_2 x_2(t) + \dots + a_m x_m(t)$$

then, by the virtue of linearity,

response $y(t)$ is given by

$$y(t) = a_1 y_1(t) + a_2 y_2(t) + \dots + a_m y_m(t)$$

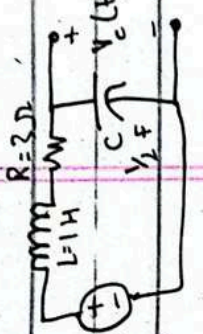
where $y_x(t)$ is the zero-state

response of the input $x_x(t)$.

⇒ If we know the system response to a unit input, we can compute the system response to any arbitrary input $x(t)$ with relative ease.

System Modelling:-

System model is a mathematical expression or a rule which satisfactorily approximate the dynamic behavior of the system.

$$V_L(t) + V_R(t) + V_C(t) = x(t)$$
$$\frac{dy}{dt} + 3y(t) + 2 \int_{-\infty}^t y(\tau) d\tau = x(t)$$


$$\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y(t) = \frac{dx}{dt} \quad (\text{differentiating})$$

$\Rightarrow D$ for the differential operator $\frac{d}{dt}$

$$(D^2 + 3D + 2)y(t) = Dx(t)$$

Total response: zero-input response, + zero-state response

$\Rightarrow$ The property of linear systems which permits the separation of an output into components resulting from initial conditions and from the input is called the decomposition property.

$\Rightarrow$ In addition to decomposition, linearity implies that both the zero input and the zero state components must obey the principle of superposition.

Example:-

Show that the system $\frac{dy}{dt} + 3y(t) = x(t)$ — ① is linear

sol. $x_1(t) \rightarrow y_1(t)$ & $x_2(t) \rightarrow y_2(t)$

$$\frac{dy_1(t)}{dt} + 3y_1(t) = x_1(t) \rightarrow \textcircled{A_1}$$

$$\frac{dy_2(t)}{dt} + 3y_2(t) = x_2(t) \rightarrow \textcircled{A_2}$$

~~$x_1(t)$~~ By $k_1 \times \textcircled{A_1}$ and $\textcircled{A_2} \times k_2$ and add them

$$\frac{d}{dt} [k_1 y_1(t) + k_2 y_2(t)] + 3[k_1 y_1(t) + k_2 y_2(t)] = k_1 x_1(t) + k_2 x_2(t)$$

(2)

When input is $k_1 x_1(t) + k_2 x_2(t)$, the system response is $k_1 y_1(t) + k_2 y_2(t)$ as shown in eq ②. So the system defined by eq ① is linear.

$\Rightarrow$ We can generalize the result to show that the system is linear when it can be represented by a differential eq. of the form

$$a_0 \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + \dots + a_n y(t)$$

$$= b_{n+m} \frac{d^m x}{dt^m} + \dots + b_{n-1} \frac{dx}{dt} + b_n x(t)$$

Zero Input Response:-

Ex: Find $y_0(t)$ the zero input component of the response for an LTIC system described by the following difference eq.

$$(D^2 + 3D + 2)y(t) = D x(t) \quad \text{--- (1)}$$

when the initial conditions are

$$y_0(0) = 0 \text{ and } \dot{y}_0(0) = -5.$$

Sol.

$y_0(t)$, the zero input component ($x(t)=0$) is solution of

$$(D^2 + 3D + 2)y_0(t) = 0 \quad \text{--- (2)}$$

Characteristic polynomial of the system is

$$\lambda^2 + 3\lambda + 2 = 0 \quad \text{--- (3)}$$

$$(\lambda + 1)(\lambda + 2) = 0 \quad \text{--- (4)}$$

The characteristic roots of the system are $\lambda_1 = -1$ and $\lambda_2 = -2$

The characteristic ~~roots~~ ^{modes} of the system are e^{-t} and e^{-2t} .

Consequently, the zero input response is

$$y_0(t) = c_1 e^{-t} + c_2 e^{-2t} \quad \text{--- (5)}$$

To find, the arbitrary constant c_1 and c_2 , we differentiate (5)

$$\dot{y}_0(t) = -c_1 e^{-t} - 2c_2 e^{-2t} \quad \text{--- (6)}$$

Setting $t=0$ in eq. (5) and (6) and substituting $y_0(0)=0$ and $\dot{y}_0(0)=-5$, we obtain

$$0 = c_1 + c_2$$

$$-5 = -c_1 - 2c_2$$

Solving these two eq. in two unknown for c_1 and c_2 yields
 $c_1 = -5$, $c_2 = 5$

$$\therefore y_0(t) = -5e^{-t} + 5e^{-2t}$$

zero-input component
of $y(t)$

Zero-State Response:-

System response $y(t)$ to an input $x(t)$ when the system is in the zero-state ~~that~~ i.e. when all initial conditions are zero.

The zero state response $y(t)$ for a linear time invariant

System

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

which is the convolution integral

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau$$

causal

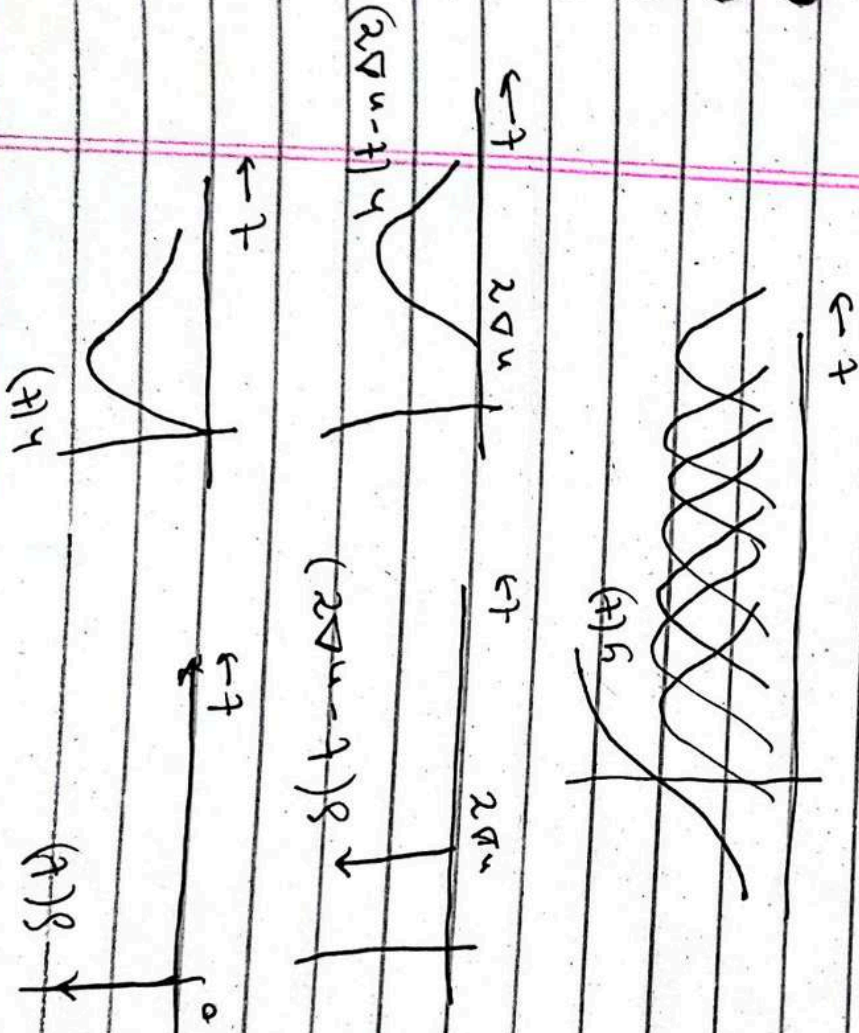
For ~~causal~~ systems, which are

LTI

$$y(t) = x(t) * h(t) = \int_{-\infty}^t x(\tau) h(t-\tau) d\tau$$

to avoid difficulty in integration if $x(t)$ contains an impulse at the origin.

There exists a characteristic mode for each characteristic root of the system and the zero state response is a linear combination of the characteristic modes of the system.



The zero-state response $y(t)$ obtained by

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

is called the convolution integral.

Convolution Integral:

$$x(t) * x_2(t) = \int_{-\infty}^{\infty} x(\tau) x_2(t - \tau) d\tau$$

System response of a LTI system

$$y(t) = \lim_{\Delta t \rightarrow 0} \sum_{n=-\infty}^{\infty} x(n\Delta t) h(t - n\Delta t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

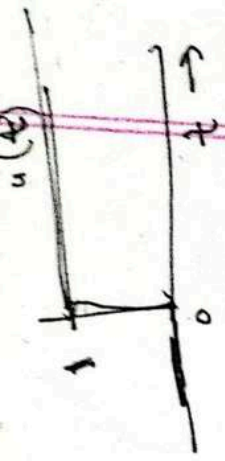
System response $y(t)$ to an arbitrary input $x(t)$ in terms of the unit impulse response $h(t)$. Knowing $h(t)$, we can determine the response $y(t)$ to any input.

$$y(x) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

works due to the all pervasive nature of the system's characteristic modes.

The system response to any input is determined by the impulse response which in turn is made of characteristic modes of the system.

So the impulse response $h(t)$ of a LTI system is readily known as the system response to an arbitrary ~~input~~ input $u(t)$ is obtained via an infinite sum of time shifted and ~~weighted~~ weighted impulse responses. This is known as convolution.



Mathematical Convolution :-

$$2 \int_0^t u(2-t)u(t)dt = (t)u * u(t) \quad (1)$$

$$2 \int_0^1 1 \cdot 1 \cdot dt =$$

$$2 \int_0^1 1 \cdot dt =$$

$$0 < t < 1 \quad t = 0 \quad \left| \begin{array}{l} 2 \\ 2 \end{array} \right|$$

$$(t)u * u(t) = t \cdot u(t)$$

$$2 \int_0^t u(2-t)u(t)dt = \int_0^t e^{at} u(t) * e^{-bt} u(t) dt \quad (2)$$

$$2 \int_0^t u(2-t)u(t)dt = \int_0^t e^{at} u(t) * e^{-bt} u(t) dt$$

$$2 \int_0^t u(2-t)u(t)dt = \int_0^t e^{at} u(t) * e^{-bt} u(t) dt$$

$$= \int_0^t e^{at} u(t) * e^{-bt} u(t) dt$$

$$= \int_0^t e^{at} u(t) * e^{-bt} u(t) dt$$

$$e^{-at} u(t) * e^{-bt} u(t) = \left(\frac{e^{-at} - e^{-bt}}{b-a} \right) u(t) \quad \text{for } t \geq 0$$

$$\frac{e^{yt-at-bt} - e^{-bt}}{e^a - e^{-b}} = e^{-at} - e^{-bt}$$

b-a Gno Badges! b-a

1, 2, 4

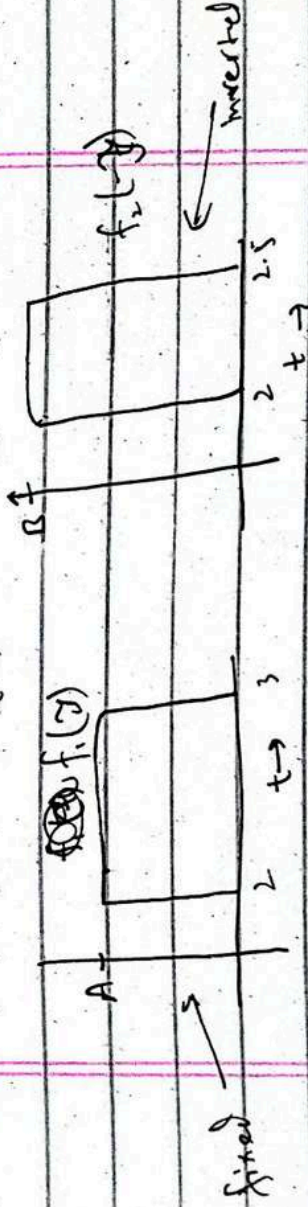
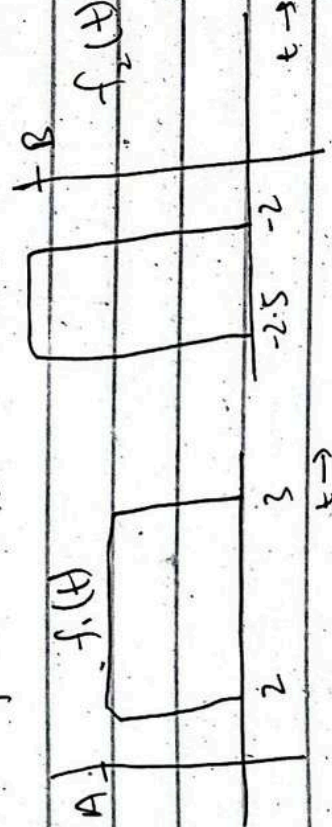
$$(1) x(t) * \delta(t - (-T)) = x(t - T)$$

$$(2) e^{\lambda t} u(t) * u(t) = \frac{1 - e^{-\lambda t}}{-\lambda} u(t)$$

$$(3) e^{\lambda_1 t} u(t) * e^{\lambda_2 t} u(t) = \frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} u(t), \lambda_1 \neq \lambda_2$$

Graphical Convolution:-

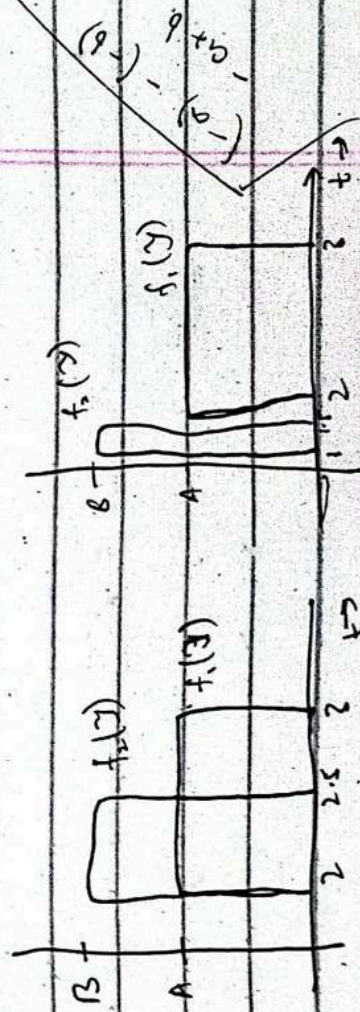
Find $f_1(t) \times f_2(t)$



$\Rightarrow$ Fix one function and invert the other one

At $t=0$

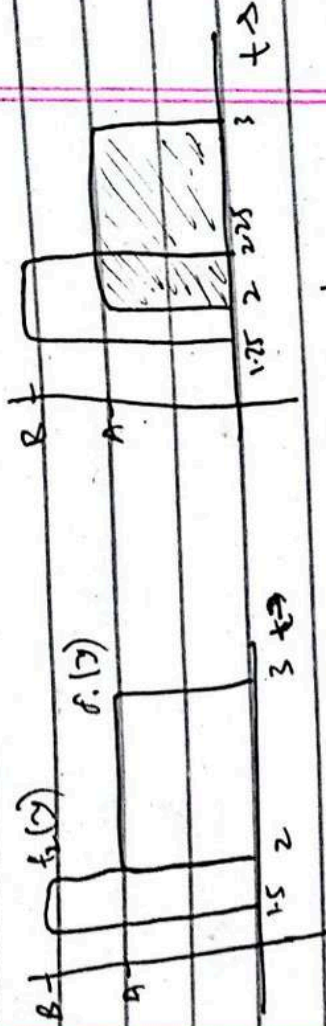
At $t=-1$



$$t = -0.25$$

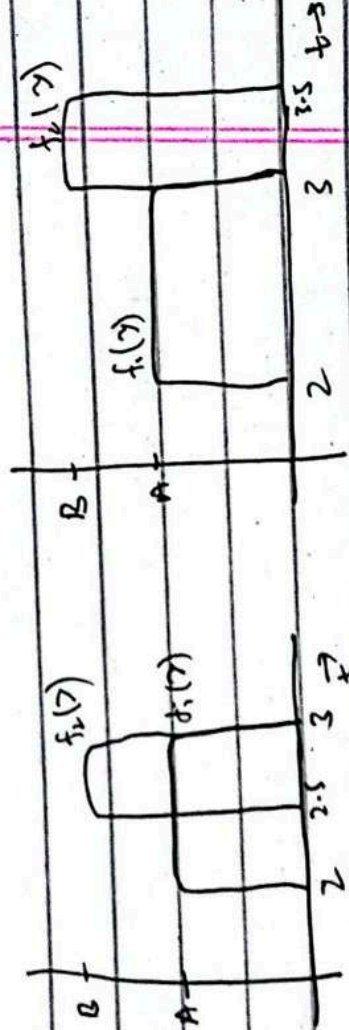
$$\text{At } \tau = -0.5$$

$$\text{At } t = 0$$

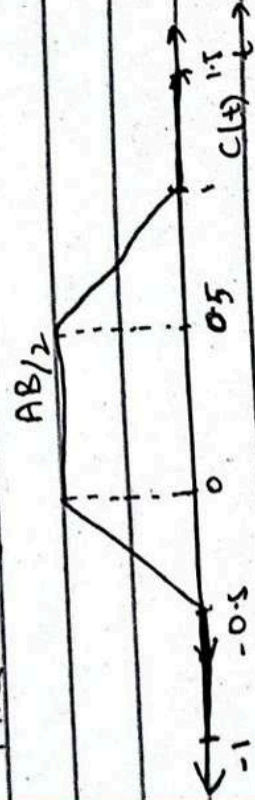


$$\text{At } t = 1$$

$$\text{At } \tau = 0.5$$



Final Convolution:-

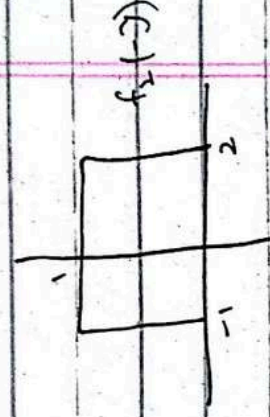
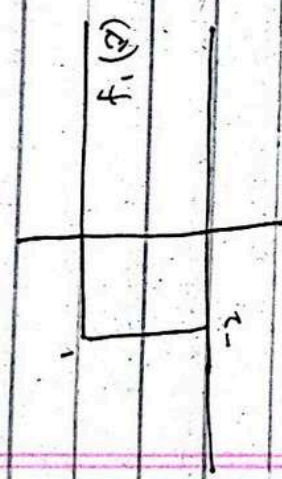
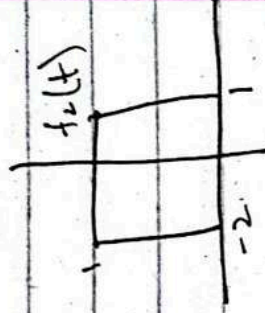
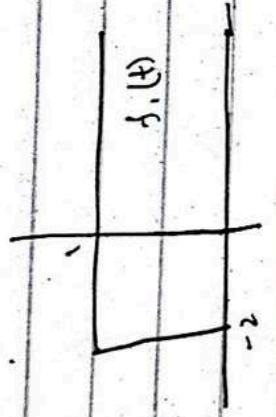


$$c(t) = AB(t + 0.5); \quad -0.5 \leq t \leq 0$$

$$c(t) = AB/2; \quad 0.5 \leq t \leq 1$$

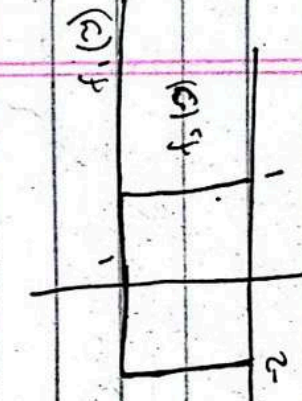
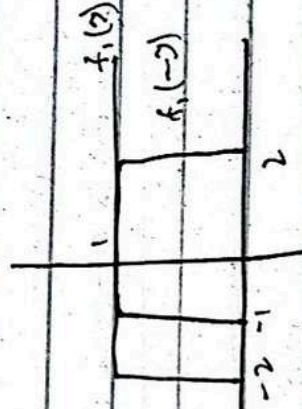
$$c(t) = 0; \quad t > 1$$

Q) $f_1(t)$ & $f_2(t)$

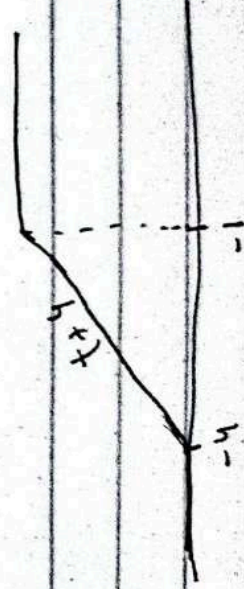
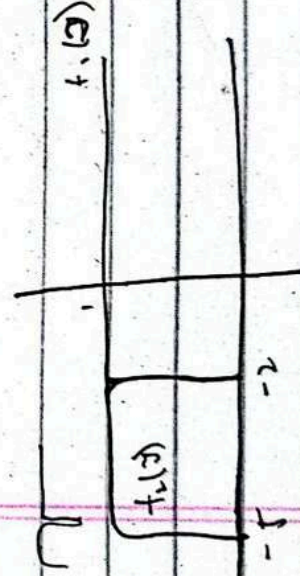


At $t=0$

At $t=-1$



At $t=-4$



$$c(t) = \begin{cases} t+4 & ; -4 \leq t \leq 1 \\ 3 & ; t > -1 \end{cases}$$

Propo Properties of Convolution:-

$$x_1(t) * x_2(t) = \int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) d\tau$$

1) Commutative Property

$$x_1(t) * x_2(t) = x_2(t) * x_1(t)$$

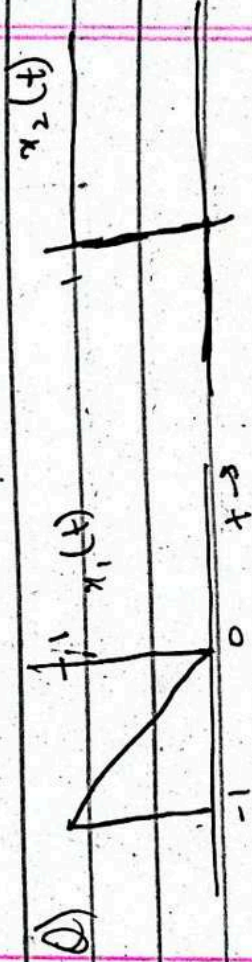
2) Distributive Property

$$x_1(t) * [x_2(t) + x_3(t)] = x_1(t) * x_2(t) + x_1(t) * x_3(t)$$

3) Associative Property

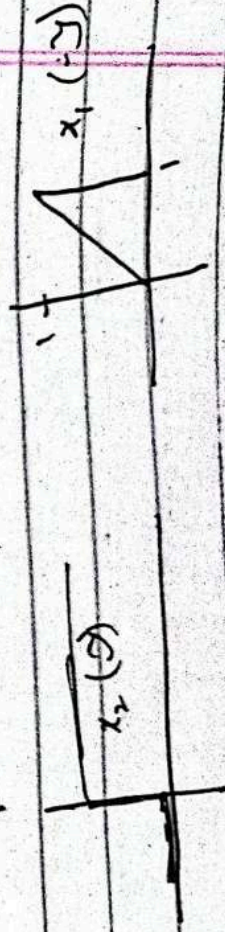
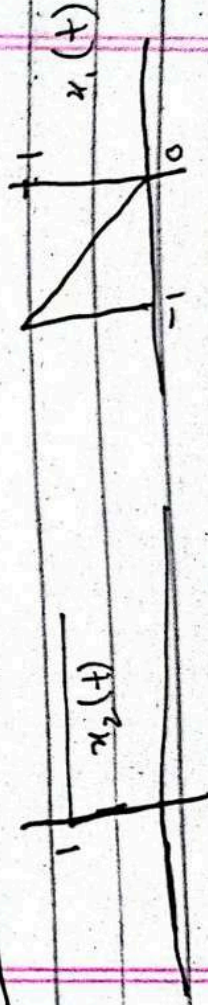
$$x_1(t) * [x_2(t) * x_3(t)] = [x_1(t) * x_2(t)] * x_3(t)$$

Graphical Convolution:-

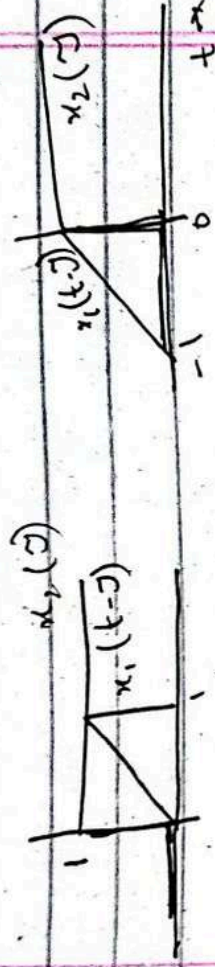


Find $x_1(t) * x_2(t)$

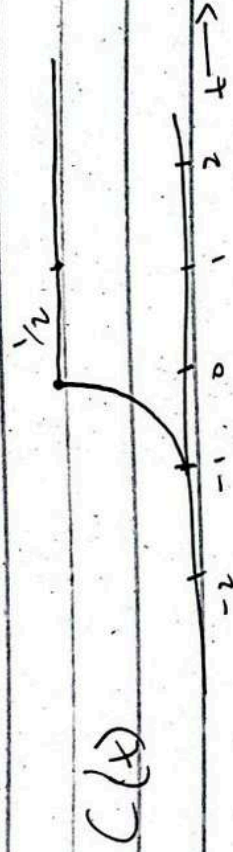
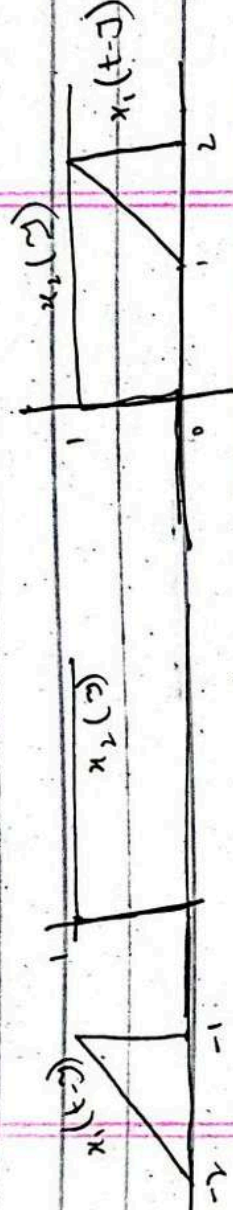
Sol. $x_1(t) * x_2(t) = x_2(t) * x_1(t)$



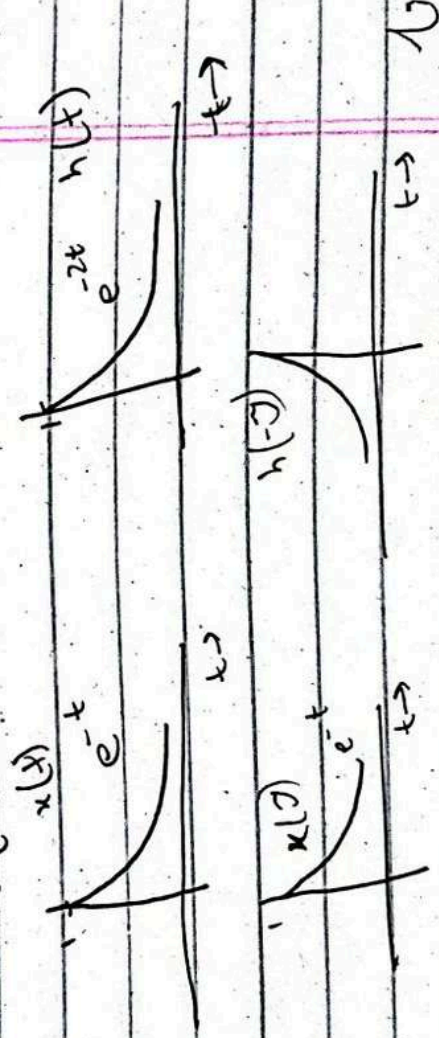
At $t=0$ At $t=-1$



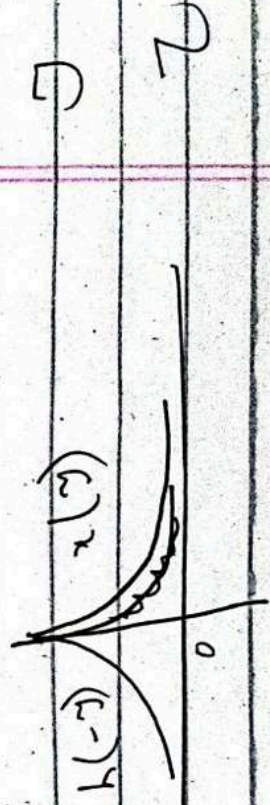
At $t=-2$



Q) Solve graphically $x(t) * h(t)$ when $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$

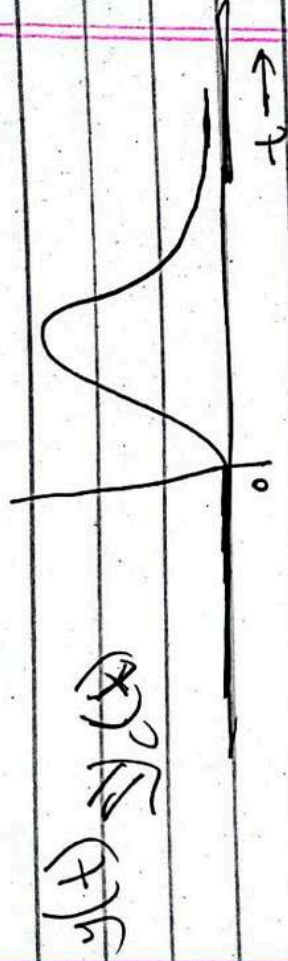
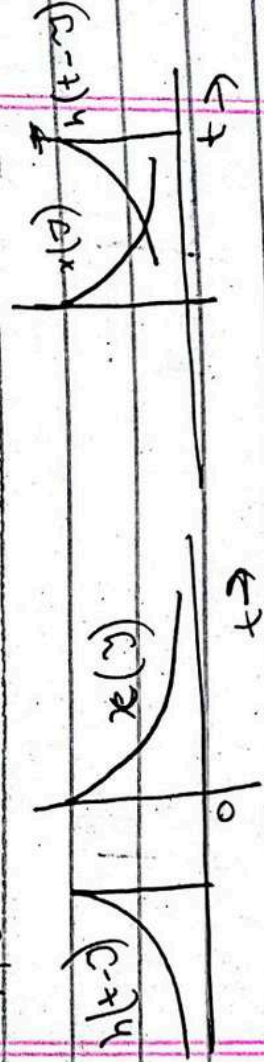


At $t=0$



At $t \leq 0$

At $t > 0$



$$y(t) = x(t) * h(t)$$

(1)

Cn #6

Fourier Series:-

Signals can be represented as a sum of its components in a variety of ways.

$$g(t) \approx c_1 x_1(t) + c_2 x_2(t) + \dots + c_N x_N(t)$$

$$= \sum_{n=1}^N c_n x_n(t) \quad ; t_1 \leq t \leq t_2$$

The Parseval's theorem says that the energy of the sum of orthogonal signal components is equal to the sum of the signal's energy.

Orthogonal Basis:-

Just as vector coordinate system can be formed by mutually orthogonal vectors (e.g. rectangular, cylindrical, spherical), we can have signal coordinate system (basis signals) formed by a variety of mutually orthogonal signals. By definition,

$$\text{Orthogonal signals} \quad \int_{-t_1}^{t_2} g(t) x(t) dt = 0$$

(2)

Two complex functions $x_1(t)$ and $x_2(t)$ are orthogonal over an interval $(t_1 \leq t \leq t_2)$

$$\int_{t_1}^{t_2} x_1(t) x_2^*(t) dt = 0$$

t_1 OR t_2

$$\int_{t_1}^{t_2} x_1^*(t) x_2(t) dt = 0$$

We can define an orthogonal signal space as

$$\int_{t_1}^{t_2} x_m(t) x_n^*(t) dt = \begin{cases} 0 & ; m \neq n \\ E_x & ; m = n \end{cases}$$

Every signal $g(t)$ can be approximated over an interval $[t_1, t_2]$ by a set of N mutually orthogonal signals

$$x_1(t), x_2(t), x_3(t), \dots, x_N(t)$$

There exists large number of orthogonal signals which can be used as basis signals for

(3)

The generalized Fourier series
e.g. ~~exponential~~ sinusoid functions,
exponential functions, Welsh functions,
Bessel functions etc.

A periodic signal $g(t)$ can
be expressed as a sum of
sinusoids of frequencies
 $0(\text{dc}), \omega_0, 2\omega_0, \dots, n\omega_0$
whose amplitudes are $a_0, a_1, a_2, \dots, a_n$

A famous set of orthogonal
signals are the sinusoidal signals
and

$$\text{[Fourier Series]} \quad g(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \quad ; t_1 \leq t \leq t_2 ; \omega_0 = \frac{2\pi}{T_0}$$

The Fourier Coefficients

a_0, a_n and b_n are

$$a_0 = \frac{1}{T} \int_{t_1}^{t_1+T_0} g(t) dt$$

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} g(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} g(t) \sin n\omega_0 t dt$$

(4)

The Fourier series can be plotted as amplitude spectrum $[C_n \text{ vs } \omega]$ or the phase spectrum $[\phi_n \text{ vs } \omega]$ which form the frequency spectra of $g(t)$.

$$\Rightarrow g(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

A signal has a dual identity time domain identity $\phi(t)$ and the frequency domain identity (Fourier spectra). Two identities complement each other.

Existence of Fourier Series (Dirichlet Conditions)

1) For the series to exist, a_0 , a_n & b_n must be finite.

Absolute Integrability

$$\int_{T_0} |g(t)| dt < \infty$$

Where Dirichlet condition, Fourier series is ~~not~~ guaranteed to exist but may not converge.

5

⇒ $g(t)$ has only finite no. of maxima and minima in one period and it may have only finite discontinuities in one period.

Combined with 1) → form strong Dirichlet conditions

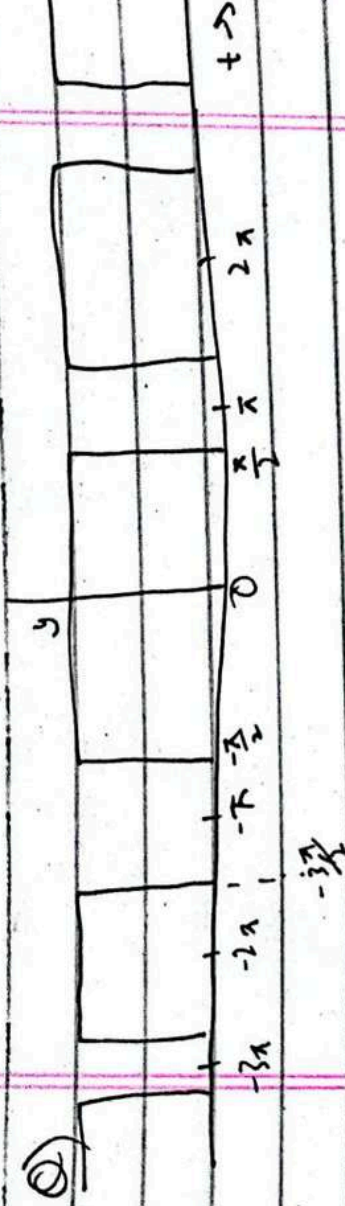
⇒ All laboratory signals satisfy both the condition and have convergent Fourier series.

Fourier Representation:-

It is a way of expressing a signal in terms of ever loading sinusoids.

Balance of magnitude and ^{phases} ~~pieces~~ of components of Fourier spectrum add up to ~~produce~~ ~~precisely~~ produce the pulse $g(t)$.

6



Find fourier series for the square pulse periodic signal.

Given $T_0 = 2\pi$ and $\omega_0 = 2\pi/T_0 = 1$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

dc-component $a_0 = \frac{1}{T_0} \int_{T_0} x(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1 dt = \frac{1}{2\pi} (t) \Big|_{-\pi}^{\pi} = \frac{1}{2}$

$$a_n = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos nt dt = \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

$$a_n = \begin{cases} 0 & ; n = \text{even} \\ \frac{2}{n\pi} & ; n = 1, 5, 9, 13, \dots \\ -\frac{2}{n\pi} & ; n = 3, 7, 11, 15, \dots \end{cases}$$

$$b_n = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \sin nt dt = 0$$

$$x(t) = \frac{1}{2} + \frac{2}{\pi} \left(\cos t - \frac{1}{3} \cos 3t + \frac{1}{5} \cos 5t - \frac{1}{7} \cos 7t + \dots \right)$$

$\frac{a+b}{a}$

7

~~cos~~ ~~sin~~

Compact Trigonometric Fourier Series:-

When $x(t)$ is real, coefficients

a_n and b_n are real for all n ,

and the Trigonometric Fourier Series

can be expressed in compact form.

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

where C_n and θ_n are related to

a_n and b_n as

$$C_0 = a_0$$

$$C_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right)$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right)$$

Periodic Signal:-

A signal $f(t)$ is said to be

periodic if for some positive

constant T_0 $f(t) = f(t + T_0)$ for all t .

The smallest value of T_0 which

satisfies $f(t) = f(t + T_0)$ is called the period of $f(t)$.

A periodic signal $f(t)$

1) must be everlasting

2) $f(t)$ can be generated by periodic extension of any segment of $f(t)$ of duration T_0 (the period).

(8)

3) Area under $f(t)$ over any interval of duration T_0 is same.

$f(t)$ can be expressed as

$$f(t) = c_0 + c_1 \cos(\omega_0 t + \theta_1) + c_2 \cos(\omega_0 t + \theta_2) + c_3 \cos(\omega_0 t + \theta_3) + \dots$$

For generality, all harmonics

up to infinity will be included.

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n)$$

$$f(t + T_0) = c_0 + \sum_{n=1}^{\infty} c_n \cos[n\omega_0(t + T_0) + \theta_n]$$

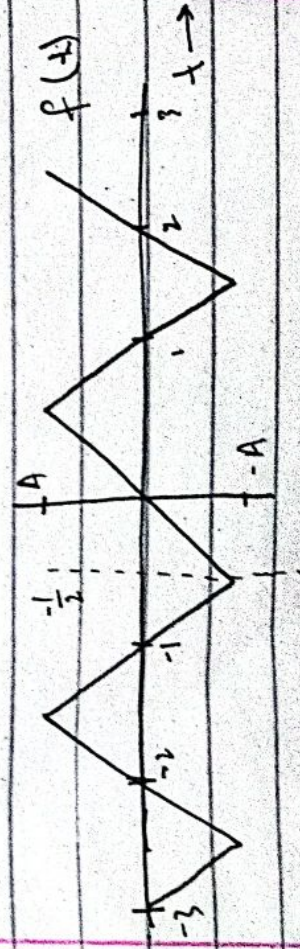
$$f(t + T_0) = c_0 + \sum_{n=1}^{\infty} c_n \cos[(n\omega_0 t + 2\pi n) + \theta_n]$$

$$= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \theta_n)$$

$$= f(t)$$

The fourier series for any even periodic function $x(t)$ consist of cosine terms only, and the series for any odd periodic function $x(t)$ consist of sin terms only.

(Q)



9

Find the compact trigonometric Fourier series for the triangular periodic signal $f(t)$ shown. Sketch the amplitude and phase spectra of $f(t)$.

$$T_0 = 2, \quad \omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{1} = \pi$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} \{ a_n \cos n\pi t + b_n \sin n\pi t \}$$

where $f(t) = \begin{cases} 2At & ; |t| < \frac{1}{2} \\ 2A(1-t) & ; \frac{1}{2} < t < \frac{3}{2} \end{cases}$

Average value (dc) of $f(t)$ is zero, $a_0 = 0$

$$a_n = \frac{2}{T_0} \int_{-T_0/2}^{T_0/2} f(t) \cos n\pi t dt$$

$$= \int_{-1/2}^{1/2} 2At \cos n\pi t dt + \int_{1/2}^{3/2} 2A(1-t) \cos n\pi t dt$$

$$= 0$$

$$b_n = \int_{-1/2}^{1/2} 2At \sin n\pi t dt + \int_{1/2}^{3/2} 2A(1-t) \sin n\pi t dt$$

$$= \frac{8A}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$f(t) = \frac{8A}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \left\{ \frac{\sin\left(\frac{n\pi}{2}\right) \sin n\pi t}{n^2} \right\}$$

(10)

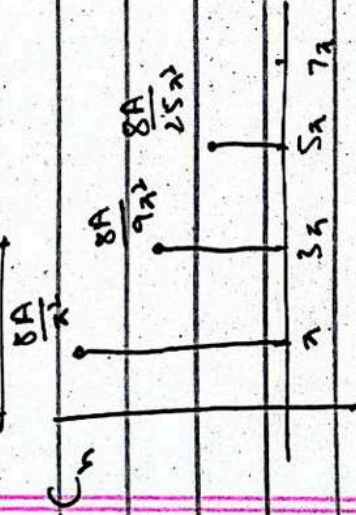
$$= \frac{8A}{\pi^2} \left[\sin \pi t - \frac{1}{9} \sin 3\pi t + \frac{1}{25} \sin 5\pi t - \frac{1}{49} \sin 7\pi t + \dots \right]$$

Identities $\sin \pi t = \cos(\pi t - 90^\circ)$

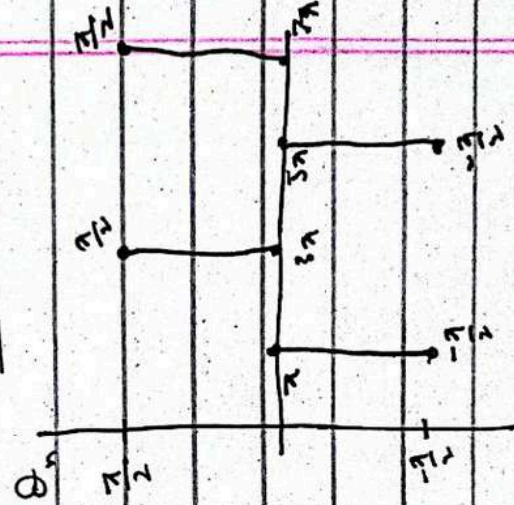
$-\sin \pi t = \cos(\pi t + 90^\circ)$

$$f(x) = \frac{8A}{\pi^2} \left[\cos(\pi t - 90^\circ) + \frac{1}{9} \cos(3\pi t + 90^\circ) + \frac{1}{25} \cos(5\pi t - 90^\circ) + \dots \right]$$

Amplitude plot



Phase Plot



11

Fourier Spectrum:-

The compact trigonometric

Fourier Series

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n (\cos n\omega_0 t + \sin n\omega_0 t) \quad \text{--- (A)}$$

indicates that a periodic signal $x(t)$

can be expressed as a sum of sinusoids

of frequencies $0, \omega_0, 2\omega_0, \dots, n\omega_0$

whose phase are $0, \theta_1, \theta_2, \dots, \theta_n$ respectively.

We can readily plot c_n versus n (amplitude spectrum)

or θ_n versus n (phase spectrum).

These plots are the frequency spectra of $x(t)$.

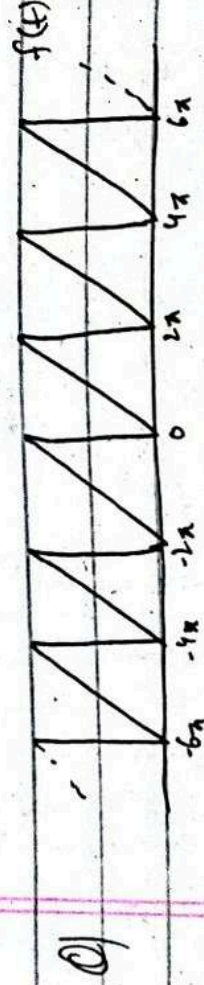
Knowing these spectra, we can reconstruct or ~~synthesize~~ synthesize the signal $x(t)$ according to eq (A).

The frequency spectra of a signal

constitute the frequency domain description

of $x(t)$ in contrast of time domain description

where $x(t)$ is specified as a function of time.



Find the trigonometric Fourier series of $f(t)$.

Sol

$$T_0 = 2\pi, \quad \omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2\pi} = 1$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$a_0 = 0.5 \text{ (by inspection)}$$

(12)

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_2} f(t) \cos n\omega_0 t dt$$

$$= \frac{1}{\pi} \int_0^{2\pi} \frac{t}{2\pi} \cos nt dt = 0$$

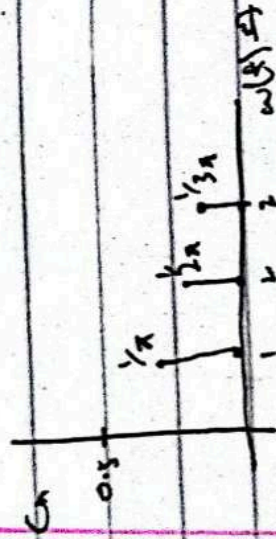
$$b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{t}{2\pi} \sin nt dt = -\frac{1}{\pi n}$$

$$f(t) = 0.5 - \frac{1}{\pi} \left[\sin t + \frac{1}{2} \sin 2t + \frac{1}{3} \sin 3t + \dots \right]$$

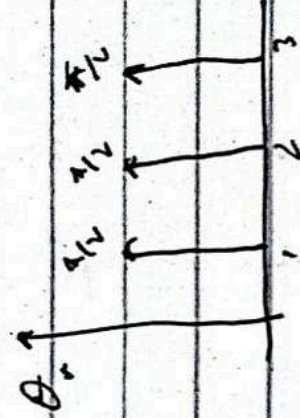
Compact Fourier Series:-

$$f(t) = 0.5 + \frac{1}{\pi} \left[\cos\left(t + \frac{\pi}{2}\right) + \frac{1}{2} \cos\left(2t + \frac{\pi}{2}\right) + \frac{1}{3} \cos\left(3t + \frac{\pi}{2}\right) + \dots \right]$$

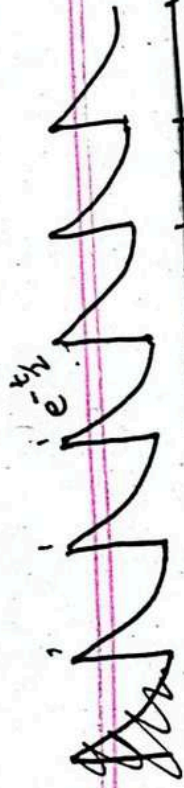
Amplitude



Phase



13



Q)

Find Compact Series

$$T_0 = \tau, \quad f_0 = \frac{1}{T_0} = \frac{1}{\tau}, \quad \omega_0 = \frac{2\pi}{T_0} \Rightarrow 2\pi f_0 \Rightarrow 2\pi \frac{1}{\tau} = \frac{2\pi}{\tau}$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos 2\pi n t / \tau + b_n \sin 2\pi n t / \tau)$$

$$a_0 = \frac{1}{\tau} \int_{-\tau}^{\tau} x(t) dt = \frac{1}{\tau} \int_0^{\tau} e^{-t/\tau} dt = 0.504$$

$$a_n = \frac{2}{\tau} \int_0^{\tau} e^{-t/\tau} \cos 2\pi n t / \tau dt$$

By using By parts

$$a_n = 0.504 \left[\frac{2}{1+16n^2} \right]$$

$$b_n = \frac{2}{\tau} \int_0^{\tau} e^{-t/\tau} \sin 2\pi n t / \tau dt = 0.504 \left[\frac{8n}{1+16n^2} \right]$$

$$x(t) = 0.504 \left[1 + \sum_{n=1}^{\infty} \frac{2}{1+16n^2} (\cos 2\pi n t / \tau + 4n \sin 2\pi n t / \tau) \right]$$

$$c_0 = a_0 = 0.504$$

$$c_n \sqrt{a_n^2 + b_n^2} = 0.504 \left[\frac{4}{(1+16n^2)^2} + \frac{64n^2}{(1+16n^2)^2} \right]$$

$$= 0.504 \left(\frac{2}{\sqrt{1+16n^2}} \right)$$

$$\theta_n = \tan^{-1} \left(\frac{-b_n}{a_n} \right) = \tan^{-1} (-4n)$$

$$= -\tan^{-1} 4n$$

$$x(t) = 0.504 + 0.504 \sum_{n=1}^{\infty} \frac{2}{\sqrt{1+16n^2}} \cos(2\pi n t / \tau - \tan^{-1} 4n)$$

(14)

Exponential Fourier Series:-

Euler's Identity $e^{j\pi} + 1 = 0$ which leads to $e^{j\omega t} = \cos \omega t + j \sin \omega t$ — (A)

A periodic signal $x(t)$ can be expressed as $\sum_{n=-\infty}^{\infty} D_n e^{jn\omega t}$ — (1)

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega t} \quad \text{--- (1)}$$

Multiply (1) with $e^{-jm\omega t}$ and integrate over one period

$$\int_{T_0} x(t) e^{-jm\omega t} dt = \sum_{n=-\infty}^{\infty} D_n \int_{T_0} e^{j(n-m)\omega t} dt \quad \text{--- (2)}$$

Orthogonality property says

$$\int_{T_0} e^{jn\omega t} e^{-jm\omega t} dt = \begin{cases} 0 & ; m \neq n \\ T_0 & ; m = n \end{cases} \quad \text{--- (3)}$$

Use (3) in (2) $D_n T_0 = \int_{T_0} x(t) e^{-jn\omega t} dt$ — (4)

$$D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega t} dt$$

To summarize the exponential

Fourier Series,

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega t}$$

$$D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega t} dt$$

Fourier Series Representation of a Periodic signal of period T_0 ($\omega_0 = \frac{2\pi}{T_0}$)

① Trigonometric series

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

② Compact Trigonometric series

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

③ Exponential series

$$f(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

Coefficient Computation:-

$$\begin{cases} a_0 = \frac{1}{T_0} \int_{T_0} f(t) dt \\ \textcircled{1} \begin{cases} a_n = \frac{2}{T_0} \int_{T_0} f(t) \cos n\omega_0 t dt \\ b_n = \frac{2}{T_0} \int_{T_0} f(t) \sin n\omega_0 t dt \end{cases} \end{cases}$$

$$\begin{cases} C_0 = a_0 \\ \textcircled{2} \begin{cases} C_n = \sqrt{a_n^2 + b_n^2} \\ \theta_n = \tan^{-1} \left(-\frac{b_n}{a_n} \right) \end{cases} \end{cases}$$

$$\textcircled{3} \begin{cases} D_n = \frac{1}{T_0} \int_{T_0} f(t) e^{-jn\omega_0 t} dt \end{cases}$$

(16)

Conversion Formulas:-

$$a_0 = c_0 = D_0$$

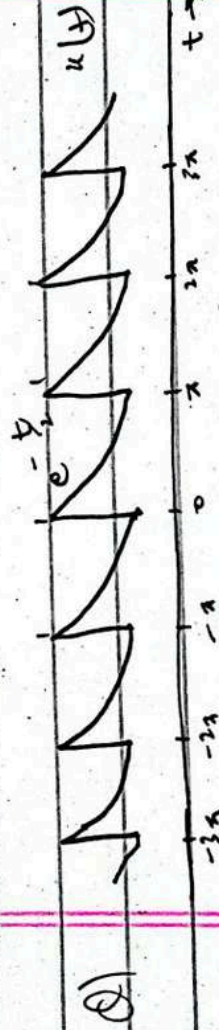
$$a_n - jb_n = c_n e^{j\theta_n} = 2D_n$$

$$a_n + jb_n = c_n e^{-j\theta_n} = 2D_n$$

$$c_0 = D_0$$

$$c_n = 2|D_n| \quad n \geq 1$$

$$\theta_n = \angle D_n$$



Find exponential Fourier series.

$$T_0 = T, \quad \omega_0 = \frac{2\pi}{T_0} = 2$$

$$u(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn2t}$$

$$\text{where } D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn2t} dt$$

$$= \frac{1}{T} \int_0^T e^{-jn2t} dt$$

$$= \frac{1}{T} \int_0^T e^{-(\frac{1}{2} + j2n)t} dt$$

$$= - \frac{1}{T(\frac{1}{2} + j2n)} e^{(\frac{1}{2} + j2n)t} \Big|_0^T$$

$$D_n = \frac{0.504}{1 + j4n}$$

(17)

$$x(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1+j4n} e^{j2nt}$$

$$x(t) = 0.504 \left[\frac{1+j4}{1+j8} e^{j4t} + \frac{1+j12}{1+j16} e^{j8t} + \dots \right. \\ \left. + \frac{1-j4}{1-j8} e^{-j4t} + \frac{1-j12}{1-j16} e^{-j8t} + \dots \right]$$

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega t} \quad \text{--- Generic Exponential Fourier Series}$$

$$x(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1+j4n} e^{j2nt} \\ = 0.504 \left[1 + \frac{1}{1+j4} e^{j2t} + \frac{1}{1+j8} e^{j4t} + \dots \right. \\ \left. + \frac{1}{1-j4} e^{-j2t} + \frac{1}{1-j8} e^{-j4t} + \dots \right]$$

Observe that the coefficients D_n are complex. Moreover, D_n and D_{-n} are conjugates.

For the exponential Fourier series, we plot the magnitude $|D_n|$ and the angle of $D_n \angle D_n$.

This requires that the coefficients D_n be expressed in polar form as $|D_n| e^{j\angle D_n}$.

(18)

For the above series,

$$D_0 = 0.504 \quad \angle D_0 = 0.122^\circ \quad \angle D_1 = -75.96^\circ \Rightarrow \angle D_1 = -75.96^\circ$$

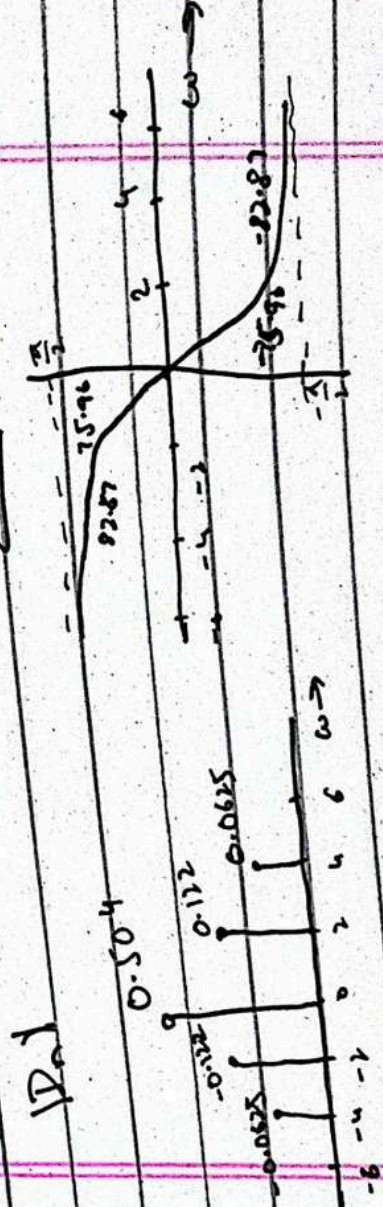
$$D_1 = \frac{0.504}{1+j4} \quad \angle D_1 = 0.122^\circ \quad \angle D_2 = -75.96^\circ$$

$$D_{-1} = \frac{0.504}{1-j4} = 0.122e \Rightarrow \angle D_{-1} = 75.96^\circ$$

$$D_2 = \frac{0.504}{1+j8} = 0.0625e \Rightarrow \angle D_2 = -82.87^\circ$$

$$D_{-2} = \frac{0.504}{1-j8} = 0.0625e \Rightarrow \angle D_{-2} = 82.87^\circ$$

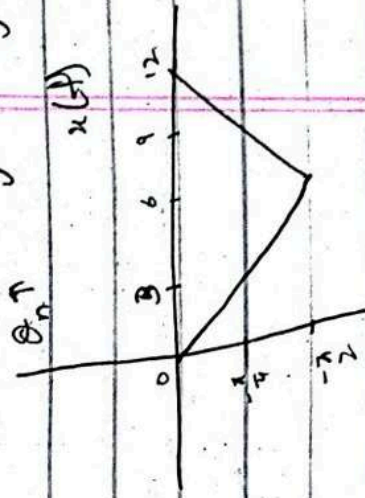
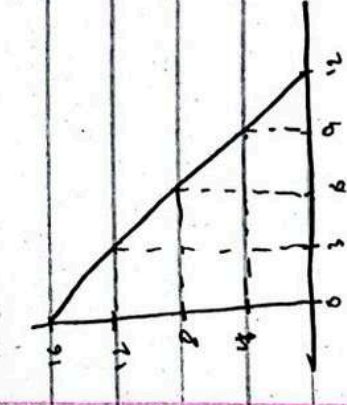
$\angle D_n$



Q.12

19

Q) The trigonometric Fourier series is shown. Sketch the exponential Fourier series and verify result analytically



The trigonometric Fourier series has four spectral components at ω frequencies 0, 3, 6, 9.

- The DC component is 16.
- Phase component of ω freq. 3 & 9 is $(-\frac{\pi}{4})$
- Phase component of ω freq. 6 is $(-\frac{\pi}{2})$

$$x(t) = 16 + 12 \cos\left(3t - \frac{\pi}{4}\right) + 8 \cos\left(6t - \frac{\pi}{2}\right) + 4 \cos\left(9t - \frac{\pi}{4}\right)$$

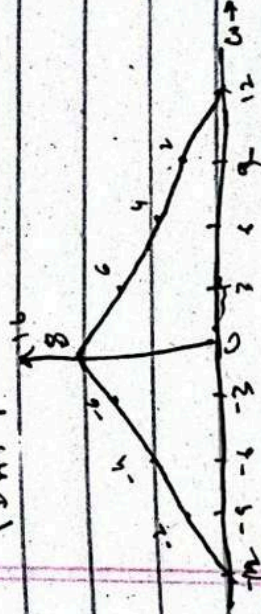
For the exponential Fourier series,

$$D_0 = C_0 = 16$$

$|D_n|$ is an even fn. of ω

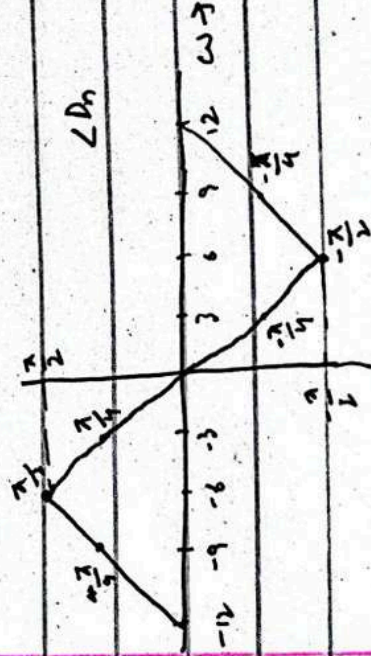
$$|D_n| = |D_{-n}| = \frac{C_n}{2}$$

$|D_n|$



(20)

$\angle D_n = \theta_n$ for $\text{five } n \neq -\theta_n$ for $\text{five } n$



$$\hat{x}(t) = 16 \times \left[6e^{-j\frac{\pi}{4}t} e^{j3t} e^{j\frac{\pi}{4}t} + 6e^{j\frac{\pi}{4}t} e^{-j3t} \right] \\ + \left[4e^{-j\frac{\pi}{4}t} e^{j6t} + 4e^{j\frac{\pi}{4}t} e^{-j6t} \right] \\ + \left[2e^{-j\frac{\pi}{4}t} e^{j9t} + 2e^{j\frac{\pi}{4}t} e^{-j9t} \right]$$

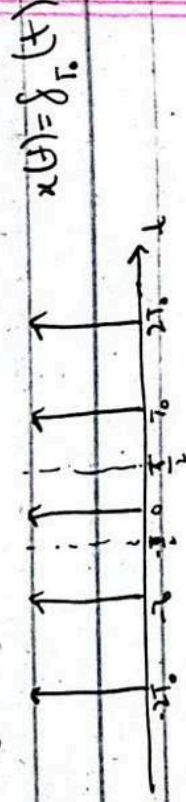
$$\hat{x}(t) = 16 \times 6 \left[e^{j(3t - \frac{\pi}{4})} + e^{-j(3t - \frac{\pi}{4})} \right] \\ + 4 \left[e^{j(6t - \frac{\pi}{4})} + e^{-j(6t - \frac{\pi}{4})} \right] \\ + 2 \left[e^{j(9t - \frac{\pi}{4})} + e^{-j(9t - \frac{\pi}{4})} \right]$$

$$x(t) = 16 + 12 \cos(3t - \frac{\pi}{4}) + 8 \cos(6t - \frac{\pi}{4}) + 4 \cos(9t - \frac{\pi}{4})$$

(21)

Impulse train and its Fourier Series:-

Impulse train is the key for analog-to-digital conversion of signals.



The unit impulse train can be expressed as

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

The exponential Fourier series for this impulse train would be

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \quad ; \omega_0 = \frac{2\pi}{T_0}$$

where

$$D_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta_{T_0}(t) e^{-jn\omega_0 t} dt$$

Over the interval $(-\frac{T_0}{2}, \frac{T_0}{2}) \rightarrow \delta_{T_0}(t) = \delta(t)$

Impulse is located at $t=0$ so using

$$\int_{-\infty}^{\infty} \phi(t) \delta(t) dt \quad [\text{sampling property}]$$

$$= \phi(0) \int_{-\infty}^{\infty} \delta(t) dt$$

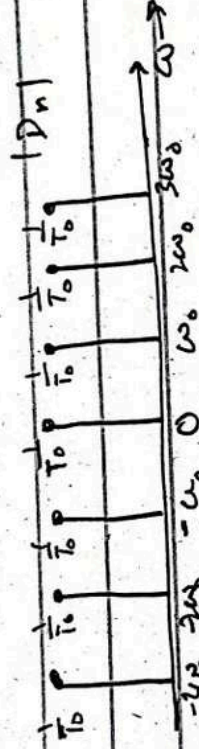
$$= \phi(0)$$

(22)

So $\int_{-\pi/2}^{\pi/2} \delta(t) e^{-jn\omega t} dt$ is evaluated at $t=0$ and becomes 1

So $D_n = \frac{1}{T_0}$

Now the exponential Fourier series is $\sum_{n=-\infty}^{\infty} \frac{1}{T_0} e^{jn\omega t}$; $\omega = \frac{2\pi}{T_0}$



is the exponential Fourier Series.

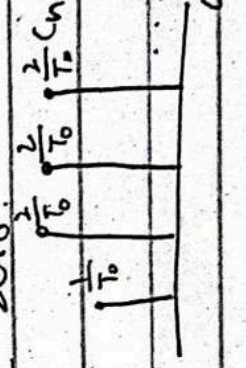
Only amplitude plot is required

As the phases are zero.

Now $C_n = D_0 = \frac{1}{T_0}$

$C_n = 2|D_n| = \frac{2}{T_0}$

$\theta_n = 0$



and $\delta_T(t) = \frac{1}{T_0} [1 + 2(\cos \omega_0 t + \cos 2\omega_0 t + \cos 3\omega_0 t + \dots)]$; $\omega_0 = \frac{2\pi}{T_0}$

is the trigonometric Fourier series for the impulse train $\delta_T(t)$

(23)

~~Cnt#1~~

Fourier Transform:-

By applying the limiting process, an aperiodic signal can be expressed as a continuous sum (integral) of overlapping exponentials.

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

is known as the Fourier Integral

$X(\omega) = F[x(t)]$ is the direct Fourier transform of $x(t)$

whereas

$x(t) = F^{-1}[X(\omega)]$ is the inverse Fourier transform of $X(\omega)$

$$x(t) \longleftrightarrow X(\omega)$$

$$\boxed{\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \end{aligned}}$$

The transform $X(\omega)$ is the frequency domain specification of $x(t)$.

The ~~spectrum~~ spectrum $X(\omega)$ can be plotted as a function of ω ~~$X(\omega)$~~

$$X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$$

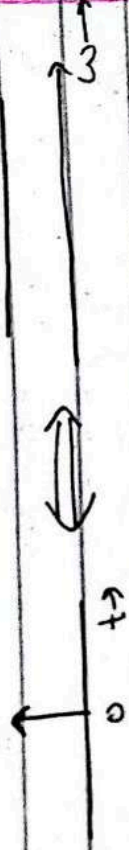
(24)

Q1) Find the Fourier transform of unit impulse function $\delta(t)$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

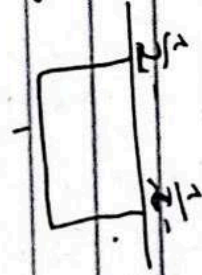
$$= 1$$



Q2) Find Fourier Transform of $g(t) = \left\{ \text{rect}\left(\frac{t}{\tau}\right) \right\}$

$$g(t) \Leftrightarrow G(\omega)$$

$$G(\omega) \Rightarrow ?$$



Sol. We have $G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\tau/2}^{\tau/2} \text{rect}\left(\frac{t}{\tau}\right) e^{-j\omega t} dt$

Since $\text{rect}\left(\frac{t}{\tau}\right) = 1$ for $|t| \leq \frac{\tau}{2}$ and

Since $\text{rect}\left(\frac{t}{\tau}\right) = 0$ for $|t| > \frac{\tau}{2}$

$$G(\omega) = \int_{-\tau/2}^{\tau/2} 1 e^{-j\omega t} dt = -\frac{1}{j\omega} \left(e^{-j\omega \frac{\tau}{2}} - e^{j\omega \frac{\tau}{2}} \right)$$

$$= 2 \sin\left(\frac{\omega \tau}{2}\right)$$

$$\omega$$

$$= \frac{\sin\left(\frac{\omega \tau}{2}\right)}{\frac{\omega \tau}{2}}$$

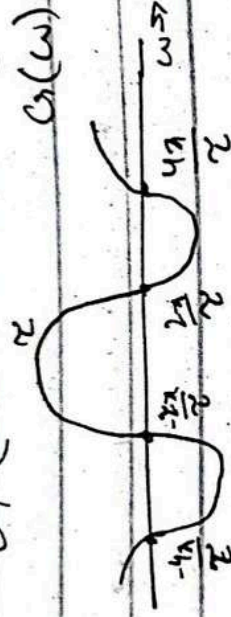
$$= \frac{\left(\frac{\tau \omega}{2}\right) \sin\left(\frac{\omega \tau}{2}\right)}{\frac{\tau \omega}{2}}$$

25

$$\left. \begin{aligned} \text{ sinc}(x) &= \frac{\sin x}{x} \\ \therefore \text{decaying sine function} \end{aligned} \right\}$$

$$= x \operatorname{sinc}\left(\frac{\omega x}{2\pi}\right)$$

$$\operatorname{rect}\left(\frac{t}{x}\right) \Leftrightarrow x \operatorname{sinc}\left(\frac{\omega x}{2\pi}\right)$$



$$Q3) \delta(t) \Leftrightarrow 1$$

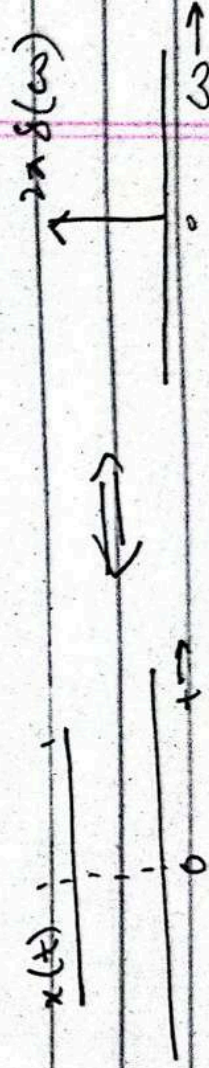
Find IFT of $\delta(\omega)$

$$f^{-1}[\delta(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} (1)$$

$$\frac{1}{2\pi} \Leftrightarrow \delta(\omega)$$

$$1 \Leftrightarrow 2\pi \delta(\omega)$$



(26)

Q4) IFT of $\delta(\omega - \omega_0)$

$$F^{-1}[\delta(\omega - \omega_0)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} e^{j\omega_0 t}$$

Therefore

$$\frac{1}{2\pi} e^{j\omega_0 t} \Leftrightarrow \delta(\omega - \omega_0)$$

$$e^{j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega - \omega_0)$$

This result shows spectrum of an everlasting exponential $e^{j\omega_0 t}$ is the single impulse at $\omega = \omega_0$.

$$\text{Ans} = e^{-j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega + \omega_0)$$

Q5) Find the FT of everlasting sinusoid $\cos \omega_0 t$.

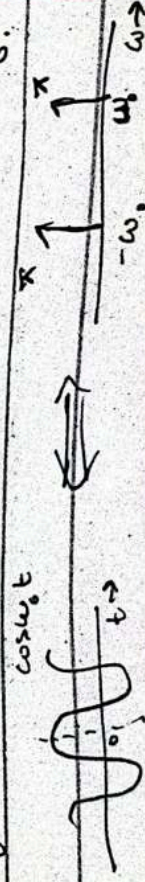
$$\cos \omega_0 t = \frac{1}{2} \left[e^{j\omega_0 t} + e^{-j\omega_0 t} \right] \int_{-\infty}^{\infty} \delta(\omega - \omega_0) + \delta(\omega + \omega_0)$$

$$e^{j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega - \omega_0)$$

$$e^{-j\omega_0 t} \Leftrightarrow 2\pi \delta(\omega + \omega_0)$$

$$\cos \omega_0 t \Leftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

The spectrum of $\cos \omega_0 t$ thus consists of two impulses at ω_0 and $-\omega_0$.



(27)

An everlasting sinusoid $\cos \omega_0 t$ can be synthesized by two everlasting exponentials $e^{j\omega_0 t}$ and $e^{-j\omega_0 t}$. Therefore, the Fourier transform of $\cos \omega_0 t$ consists of only two components of frequencies ω_0 & $-\omega_0$.

Q6) FT of $e^{-at} u(t)$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= -\frac{1}{a+j\omega} \left[e^{-(a+j\omega)t} \right]_0^{\infty}$$

But $|e^{-j\omega t}| = 1$

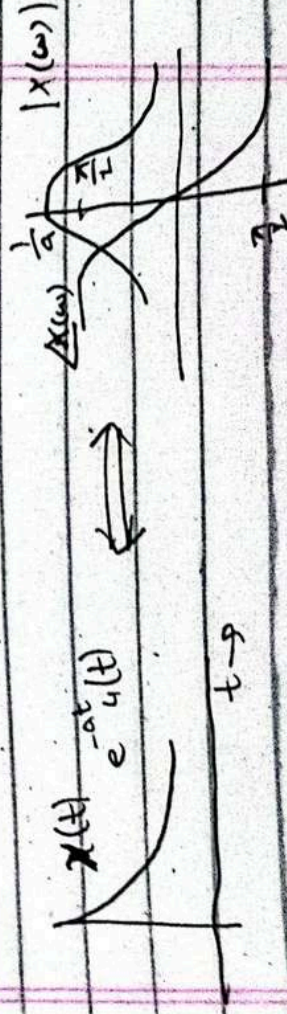
If $a < 0$, as $t \rightarrow \infty$, $e^{-(a+j\omega)t} = e^{-at} e^{-j\omega t} = \infty$ [Growing exponentially]

If $a > 0$, $X(\omega) = \frac{1}{a+j\omega}$

Expressing $\frac{1}{a+j\omega}$ in polar form as $\frac{1}{\sqrt{a^2+\omega^2}} e^{-j \tan^{-1}(\frac{\omega}{a})}$

$$X(\omega) = \frac{1}{\sqrt{a^2+\omega^2}} e^{-j \tan^{-1}(\frac{\omega}{a})}$$

$$|X(\omega)| = \frac{1}{\sqrt{a^2+\omega^2}} \quad \angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$



Properties of Fourier Transform

1) Linearity

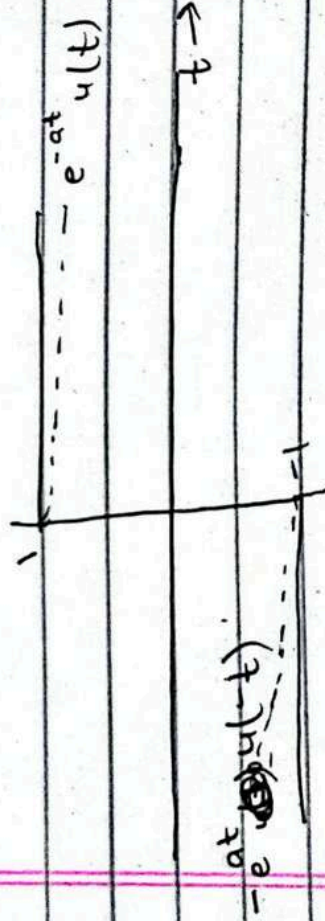
$$x_1(t) \Leftrightarrow X_1(\omega)$$

$$x_2(t) \Leftrightarrow X_2(\omega)$$

$$a_1 x_1(t) + a_2 x_2(t) \Leftrightarrow a_1 X_1(\omega) + a_2 X_2(\omega)$$

Q) Find FT for $\text{sgn } t$ (signum t)

$$\text{sgn } t = \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$$



$$\text{sgn } t = \lim_{a \rightarrow 0} \left[e^{-at} u(t) - e^{at} u(-t) \right]$$

$$F\{\text{sgn } t\} = \lim_{a \rightarrow 0} \left[F\{e^{-at} u(t)\} - F\{e^{at} u(-t)\} \right]$$

$$= \lim_{a \rightarrow 0} \left[\frac{1}{a+j\omega} - \frac{-1}{a-j\omega} \right]$$

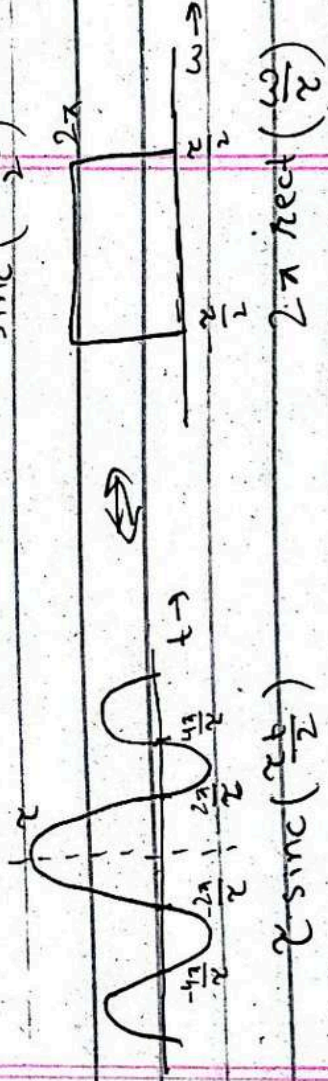
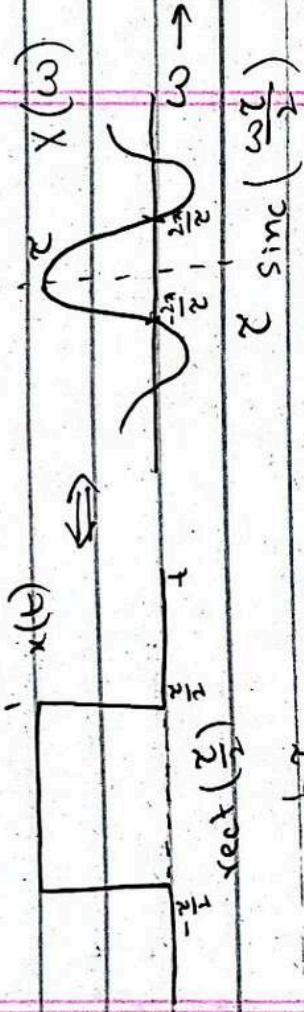
$$= \lim_{a \rightarrow 0} \left[\frac{-2j\omega}{a^2 + \omega^2} \right]$$

$$= \frac{2}{j\omega}$$

2) Duality

$$x(t) \Leftrightarrow X(\omega)$$

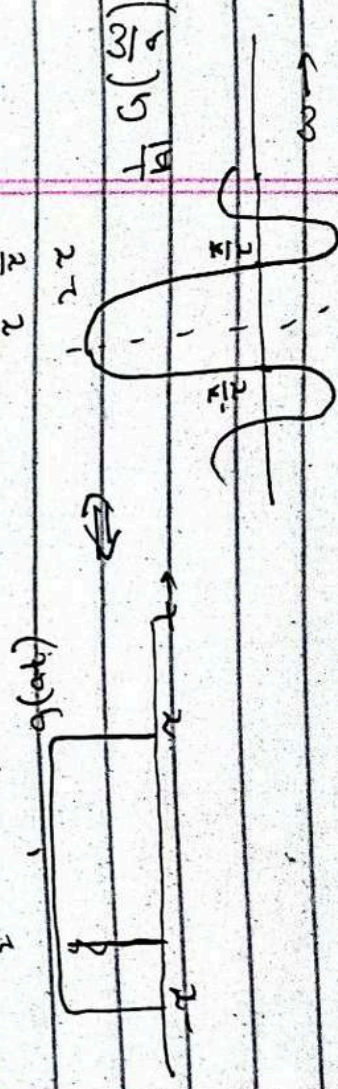
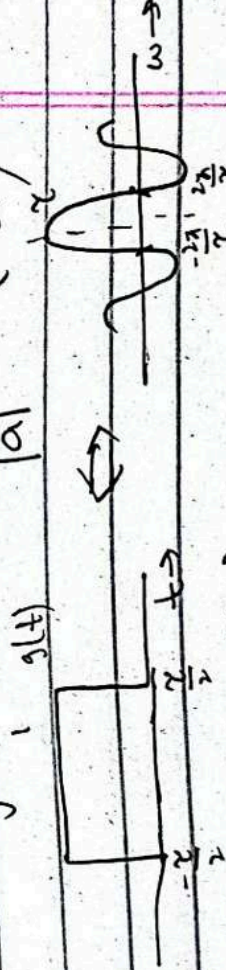
$$X(t) \Leftrightarrow 2\pi x(-\omega)$$



3) Scaling

$$g(t) \Leftrightarrow G(\omega)$$

$$g(at) \Leftrightarrow \frac{1}{|a|} G\left(\frac{\omega}{a}\right)$$



-) The scaling property states that time compression of a signal results in the spectral expansion of the signal and time expansion of a signal results in spectral compression.

4) Time Shifting Property

$$g(t) \Leftrightarrow G(\omega)$$

$$g(t - t_0) \Leftrightarrow G(\omega) e^{-j\omega t_0}$$

Property states that delaying a signal by " t_0 " seconds does not change its amplitude spectrum. The phase spectrum is shifted by ωt_0 .

Fourier Transforms:-

$$x(t) \quad X(\omega)$$

$$e^{-at} u(t) \quad \frac{1}{a + j\omega}$$

$$e^{at} x(-t) \quad \frac{1}{a - j\omega}$$

$$e^{-a|t|} \quad \frac{2a}{a^2 + \omega^2}$$

$$\delta(t) \quad 1$$

$$1 \quad 2\pi \delta(\omega)$$

$$e^{j\omega_0 t}$$

$$2\pi \delta(\omega - \omega_0)$$

$$\cos \omega_0 t \quad \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$\sin \omega_0 t \quad j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$

$$\text{sgn} t \quad \frac{2}{j\omega}$$

$$\frac{\omega}{\pi} \text{sinc}(\omega t) \quad \text{rect}\left(\frac{\omega}{2\omega_0}\right)$$

4.5) Frequency Shifting Property (ω_0 real)

$$g(t) e^{j\omega_0 t} \Leftrightarrow G(\omega) \Leftrightarrow G(\omega - \omega_0) \Leftrightarrow G(\omega + \omega_0)$$

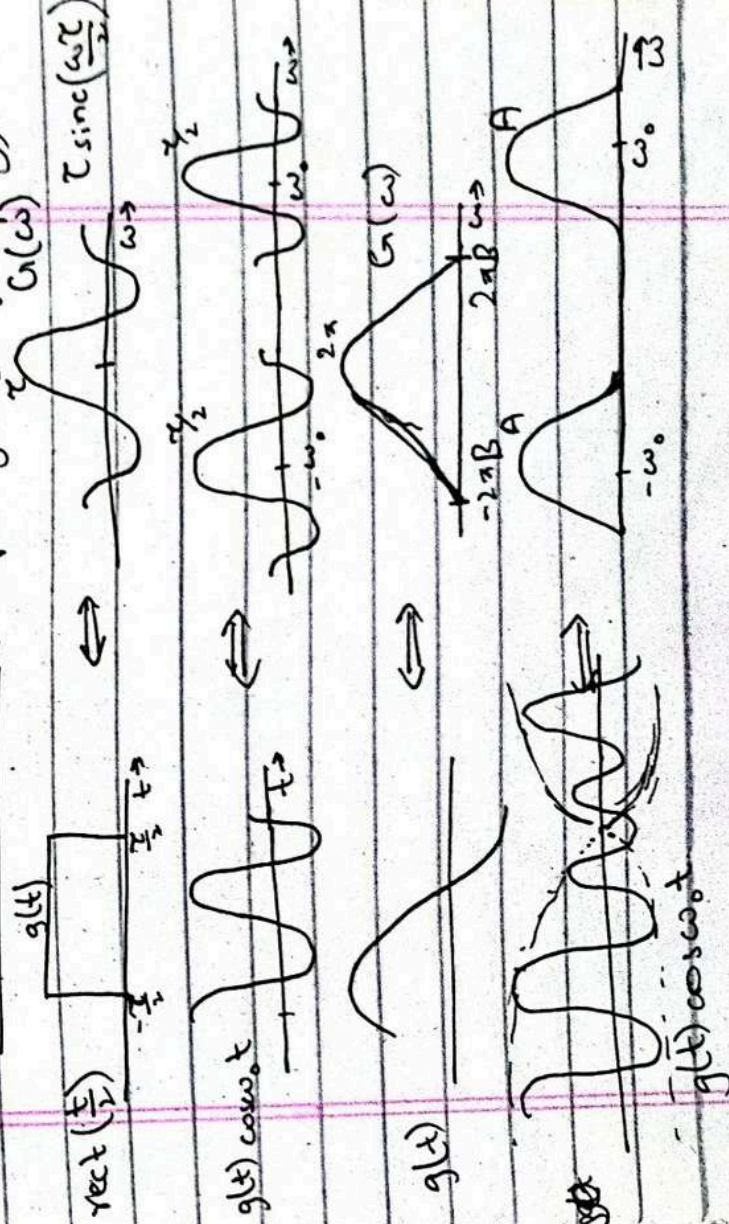
Since $e^{j\omega_0 t}$ is not a real funcⁿ, frequency shifting is achieved by multiplying $g(t)$ with a sinusoid

$$g(t) - \cos\omega_0 t = \frac{1}{2} [g(t) e^{j\omega_0 t} + g(t) e^{-j\omega_0 t}]$$

$$g(t) \cos\omega_0 t \Leftrightarrow \frac{1}{2} [G(\omega - \omega_0) + G(\omega + \omega_0)]$$

The final result shows that multiplication a signal $g(t)$ by a sinusoid of ω_0 , shifts the spectrum $G(\omega)$ by $\pm\omega_0$.

Fourier Transforms (Time Shifting and Frequency Shifting Property)



→ Convolution $g(t) * x(t) = \int_{-\infty}^{\infty} g(\tau) x(t-\tau) d\tau$

6) Time Convolution property

$$g_1(t) * g_2(t) \Leftrightarrow G_1(\omega) G_2(\omega)$$

If $g_1(t) \Leftrightarrow G_1(\omega)$ and $g_2(t) \Leftrightarrow G_2(\omega)$

then time convolution $g_1(t) * g_2(t) \Leftrightarrow G_1(\omega) G_2(\omega)$

(Frequency convolution) $g(t) g_2(t) \Leftrightarrow \frac{1}{2\pi} G_1(\omega) G_2(\omega)$

Proof of Time Convolution: $F[g(t) * g_2(t)]$

$$= \int_{-\infty}^{\infty} e^{-j\omega t} \left[\int_{-\infty}^{\infty} g(\tau) g_2(t-\tau) d\tau \right] dt$$

∴ Changing order of integration

$$= \int_{-\infty}^{\infty} g_1(\tau) \left[\int_{-\infty}^{\infty} e^{-j\omega t} g_2(t-\tau) dt \right] d\tau$$

$$= \int_{-\infty}^{\infty} g_1(\tau) G_2(\omega) e^{-j\omega \tau} d\tau$$

$$= G_1(\omega) \int_{-\infty}^{\infty} e^{-j\omega \tau} g_1(\tau) d\tau$$

$$= G_1(\omega) G_2(\omega)$$

$$= G_1(\omega) G_2(\omega)$$

Time Differentiation Property

$$x(t) \Leftrightarrow X(\omega)$$

$\Rightarrow$ then $\frac{dx}{dt} \Leftrightarrow j\omega X(\omega)$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Differentiate both sides

$$\frac{dx}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} j\omega X(\omega) e^{j\omega t} d\omega$$

$$\frac{dx}{dt} \Leftrightarrow j\omega X(\omega)$$

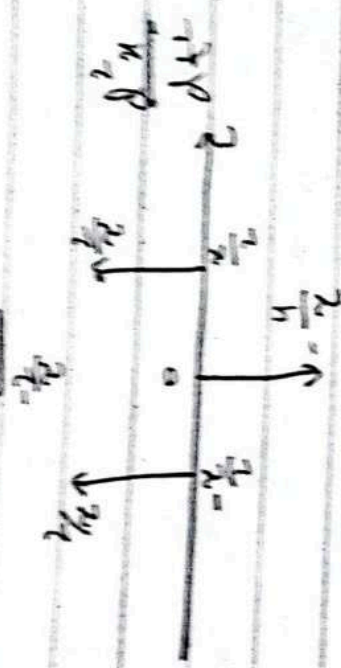
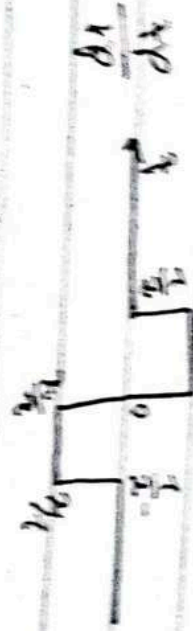
Repeated application of differentiation yields:

$$\frac{d^n x}{dt^n} \Leftrightarrow (j\omega)^n X(\omega)$$

8) Time Integration Property

$$\int_{-\infty}^{\infty} x(\tau) d\tau \Leftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0) \delta(\omega)$$

Ex) Find the FT of triangles f_n.
 $x(t) = \Delta(\frac{t}{\tau})$



From time shifting property $\delta(t-t_0) \rightarrow e^{-j\omega t_0}$

$$\frac{d^2x}{dt^2} = \frac{1}{\tau} \delta\left(t + \frac{\tau}{2}\right) - \frac{1}{\tau} \delta\left(t - \frac{\tau}{2}\right)$$

Taking Fourier transform

$$-\omega^2 X(\omega) = \frac{1}{\tau} \left[e^{j\omega \frac{\tau}{2}} - e^{-j\omega \frac{\tau}{2}} \right]$$

$$= \frac{1}{\tau} (1 - \cos \omega \tau)$$

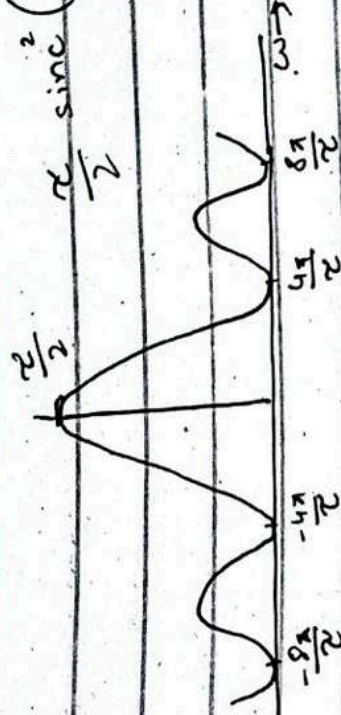
$$-\omega^2 X(\omega) = \frac{1}{\tau} (1 - \cos \omega \tau)$$

and

$$X(\omega) = \frac{1}{\omega^2 \tau} (1 - \cos \omega \tau)$$

$$= \frac{1}{\omega^2 \tau} \left(\frac{\omega \tau}{2} \right)^2 \sin^2 \left(\frac{\omega \tau}{2} \right)$$

$$\frac{x}{2} \quad x \operatorname{sinc}^2\left(\frac{\omega x}{4}\right)$$



⇒ This procedure can be applied to any function made up of straight line segments with $x(t) \rightarrow 0$ as $|t| \rightarrow \infty$.

Fourier Transform Operations

$$x(t) \quad X(\omega)$$

$$1) \text{ Scalar Multiplication } kx(t) \quad kX(\omega)$$

$$2) \text{ Addition } x_1(t) + x_2(t) \quad X_1(\omega) + X_2(\omega)$$

$$3) \text{ Duality } X(t) \quad 2\pi x(-\omega)$$

$$4) \text{ Scaling } x(at) \quad \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$5) \text{ Time Shifting } x(t-t_0) \quad X(\omega) e^{-j\omega t_0}$$

$$6) \text{ Frequency Shifting } x(t) e^{j\omega_0 t} \quad X(\omega - \omega_0)$$

$$x(t) \cos \omega_0 t = \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

$$7) \text{ Time convolution } x_1(t) * x_2(t) \quad X_1(-\omega) X_2(\omega)$$

$$8) \text{ Frequency Convolution } x_1(t) x_2(t) \quad \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

Q1) Find FT of

$$x(t) = e^{-|t|} \cos(2t)$$

$$e^{-at} u(t) \Leftrightarrow \frac{1}{a+j\omega}$$

$$e^{-t} u(t) \Leftrightarrow \frac{1}{1+j\omega}$$

$$e^{-a|t|} \Leftrightarrow \frac{2a}{a^2 + \omega^2}$$

$$e^{-|t|} \Leftrightarrow \frac{2}{1+\omega^2}$$

$$\cos \omega_0 t \Leftrightarrow \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

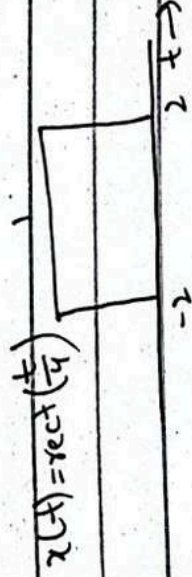
$$\cos 2t \Leftrightarrow \pi [\delta(\omega - 2) + \delta(\omega + 2)]$$

Freq. Shift $x(t) \cos \omega_0 t \Leftrightarrow \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$

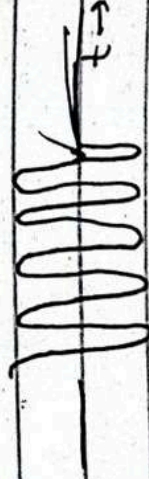
$$\Rightarrow e^{-|t|} \cos 2t \Leftrightarrow \frac{1}{1+(\omega-2)^2} + \frac{1}{1+(\omega+2)^2}$$

Q2) Find FT of $x(t) \cos 10t$ in which

$x(t)$ is a gate pulse $\text{rect}(\frac{t}{4})$



$$x(t) \cos 10t$$



Sol: $x(t) = \text{rect}(\frac{t}{4}) \Leftrightarrow X(\omega) = 4 \text{sinc}(2\omega)$

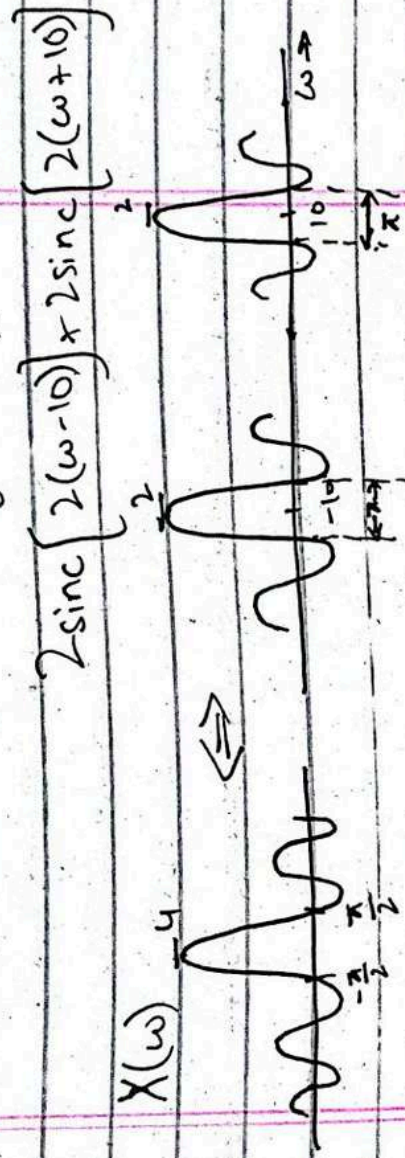
$$x(t) \cos 10t \Leftrightarrow ?$$

Freq Shifting $x(t) \cos \omega_0 t \Leftrightarrow \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$

$$x(t) \cos 10t \Leftrightarrow \frac{1}{2} [X(\omega - 10) + X(\omega + 10)]$$

$$X(\omega) = 4 \operatorname{sinc}(2\omega)$$

$$\operatorname{rect}\left(\frac{t}{4}\right) \Leftrightarrow 4 \operatorname{sinc}(2\omega) \quad \text{or} \quad 2 \operatorname{sinc}(\omega - 10) + 2 \operatorname{sinc}(\omega + 10)$$



Q) Find the FT of signal $x(t) = \frac{2}{1+t^2}$

$$e^{-|t|} \Leftrightarrow \frac{2}{1+\omega^2}$$

$\Rightarrow$ Duality Property

$$x(t) \Leftrightarrow X(\omega)$$

$$X(t) \Leftrightarrow 2\pi x(-\omega)$$

$$e^{-|t|} \Leftrightarrow \frac{2}{1+\omega^2}$$

Using duality property

$$x(t) \Leftrightarrow X(\omega)$$

$$e^{-|t|} \Leftrightarrow \frac{2}{1+\omega^2}$$

$$X(t) \Leftrightarrow 2\pi x(-\omega)$$

$$e^{-\frac{2}{1+t^2}} \Leftrightarrow 2\pi e^{-|-\omega|}$$

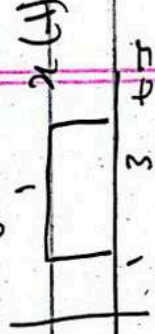
Q1) Find FT of $x \text{ sinc}\left(\frac{x}{2}\right)$

$$\underbrace{\text{rect}\left(\frac{t}{2}\right)}_{x(t)} \Leftrightarrow \underbrace{2 \text{ sinc}\left(\frac{\omega x}{2}\right)}_{X(\omega)}$$

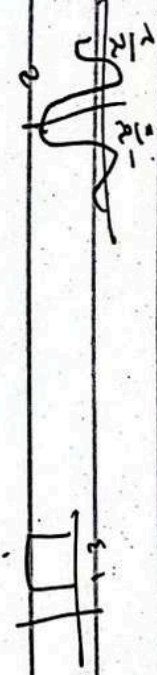
$$\underbrace{2 \text{ sinc}\left(\frac{x}{2}\right)}_{X(t)} \Leftrightarrow \underbrace{2\pi \text{ rect}\left(-\frac{\omega}{2}\right)}_{2\pi X(-\omega)}$$

Q2) Find FT of following signal

1) By definition
2) By properties



$$\text{rect}\left(\frac{t}{2}\right) \Leftrightarrow 2 \sin\left(\frac{\omega x}{2}\right)$$



By definition

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad x(t) = \text{rect}\left(\frac{t}{2}\right) = 1$$

$$= \int_{-1}^1 e^{-j\omega t} dt$$

$$= -\frac{1}{j\omega} \left[e^{-j\omega} - e^{j\omega} \right]$$

$$= \frac{e^{-j\omega} - e^{j\omega}}{j\omega} \left[e^{j\omega} - e^{-j\omega} \right]$$

$$= \frac{e^{-j\omega} - e^{j\omega}}{\omega} \left[\frac{e^{j\omega} - e^{-j\omega}}{j} \right]$$

$$= e^{-j2\omega} \left(\frac{2 \sin \omega}{\omega} \right)$$

$$= 2 e^{-j2\omega} \operatorname{sinc} \left(\frac{\omega}{2} \right)$$

$$F(\omega) = 2 e^{-j2\omega} \operatorname{sinc}(\omega)$$

DRG by 8148.

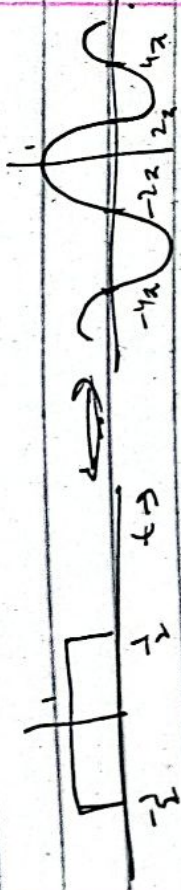
$$\operatorname{rect}\left(\frac{t}{2}\right) \Leftrightarrow 2 \operatorname{sinc} \frac{\omega}{2}$$

$$\operatorname{rect}\left(\frac{t}{2}\right) \Leftrightarrow 2 \operatorname{sinc} \omega$$

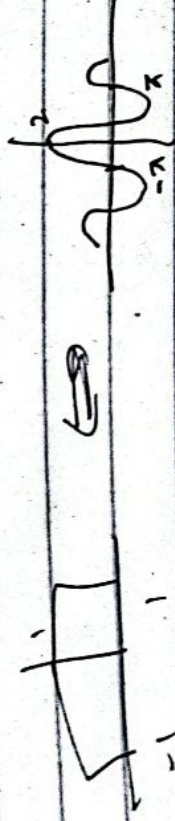
Time shifting property
 $x(t-t_0) \Leftrightarrow X(\omega) e^{-j\omega t_0}$

$$\operatorname{rect}\left(\frac{t-2}{2}\right) \Leftrightarrow 2 e^{-j2\omega} 2 \operatorname{sinc}(\omega)$$

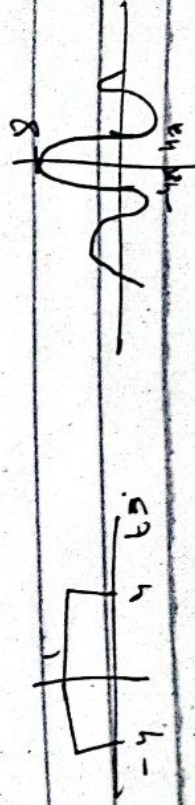
$$\operatorname{rect}(t) \Leftrightarrow \operatorname{sinc}\left(\frac{\omega}{2}\right)$$



$$\operatorname{rect}\left(\frac{t}{2}\right) \Leftrightarrow 2 \operatorname{sinc}(\omega)$$



$$\operatorname{rect}\left(\frac{t}{8}\right) \Leftrightarrow 8 \operatorname{sinc}\left(\frac{\omega}{4}\right)$$



Diff. 15V
GND
CIRCUIT
CLK

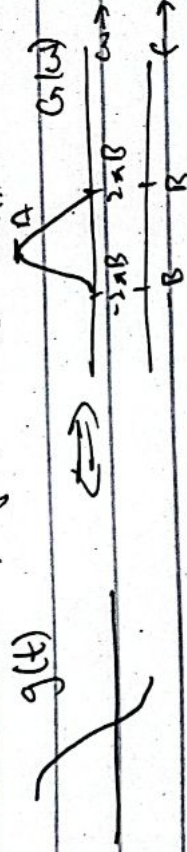
Sampling Theorem:-

Any signal whose spectrum is bandlimited to

B Hz $[G(\omega) = 0 \text{ for } |\omega| > 2\pi B]$

can be reconstructed exactly (without errors) from its samples taken uniformly at a rate $R > 2B$ samples/second.

Min. sampling frequency $f_{s, \min} = 2B$ Hz



⇒ Multiplication with an impulse train

$$g_{Ts}(t) = g(t) \delta_{Ts}(t) = \frac{1}{T_s} [1 + 2\cos\omega_s t + 2\cos 2\omega_s t + 2\cos 3\omega_s t + \dots]$$

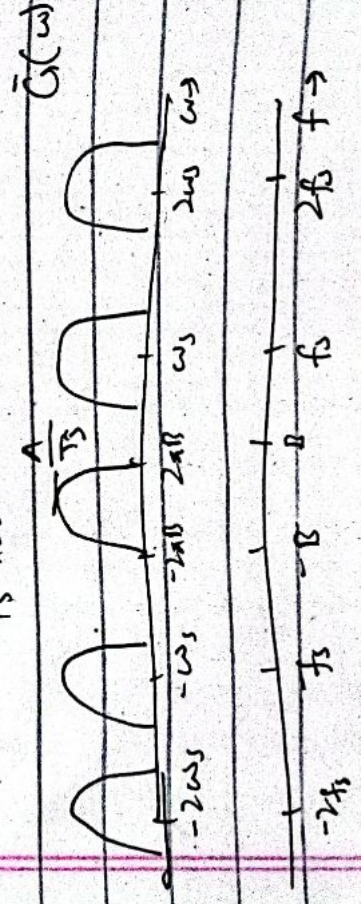
$$\omega_s = \frac{2\pi}{T_s} = 2\pi f_s$$

$$\bar{g}(t) = g(t) \delta_{Ts}(t)$$

$$\bar{g}(t) = \frac{1}{T_s} [g(t) + 2g(t)\cos\omega_s t + 2g(t)\cos 2\omega_s t + \dots]$$

$$G(\omega) \left(+2g(t)\cos 3\omega_s t + \dots \right) \rightarrow G(\omega - 2\omega_s) + G(\omega + 2\omega_s)$$

$$\bar{G}(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(\omega - n\omega_s)$$



PCM = Pulse Code Modulation

To reconstruct $g(t)$ from $\bar{g}(t)$, $G(\omega)$ needs to be recovered from $\bar{G}(\omega)$. This is only possible when there is ~~no~~ no overlap b/w successive cycles of $\bar{G}(\omega)$, so

$$f_s > 2B \quad \text{or} \quad \left[T_s < \frac{1}{2B} \right]$$

⇒ The min. sampling frequency $f_s = 2B$ required to recover $g(t)$ from its sample $\bar{G}(t)$ is called Nyquist rate for $g(t)$ and the corresponding sampling interval $T_s = \frac{1}{2B}$ is called Nyquist interval.

Q) A television signal, audio & video, has a bandwidth of 4.5 MHz. This signal is sampled, quantized and binary coded to obtain a PCM signal.

a) Determine sampling rate if signal is sampled at 20% above the Nyquist rate.

Sol- $B = 4.5 \text{ MHz}$
sample, quantized and binary coded

✓ Nyquist rate = $2B = 2 \times 4.5 = 9 \text{ MHz}$

100% above
20% above
NS = 120%
Sampling rate = 120% of 9.5 MHz = 10.8 M sample/sec

b) If sample are quantized into 1024 levels, determine the binary pulses required to encode each sample

Sol- 1024 levels $0_0 - 0_{1023}$

$$2^{10} = 1024$$

10 binary pulses to encode each sample.

